MULTI-HEAD ATTENTION TRANSFORMER FOR FILIPINO SIGN LANGUAGE

A Thesis
Presented to the Faculty of the
College of Computer and Information Sciences
Polytechnic University of the Philippines
Sta. Mesa, Manila

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Computer Science

by

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July 2025

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

I am Novelle Lyn Estrella, faculty member/student/administrative employee of the Polytechnic University of the Philippines here declare that I HAVE /HAVE NOT used Artificial Intelligence to complete this research project/thesis/dissertation titled Multi-Head Attention Transformer for Filipino Sign Language.

The use of Artificial Intelligence is in the following manner:

AI Tool Used	Detailed Prompts and AI Outputs	How the output was incorporated in the project
ChatGPT	Prompt: “Assume the role of a computer science thesis panel member evaluating the attached thesis proposal draft (Chapters 1-3). Conduct a thorough assessment, identifying weaknesses, errors, and missing information in each section. Provide specific, actionable comments and suggestions for improvement, ensuring clarity, coherence, and alignment with academic standards. Since this draft does not yet include experimentations, focus on evaluating the logical consistency of the research framework, methodology, and problem statement. Make your feedback as precise and detailed as possible, offering guidance on how to strengthen the proposal for further development.” Output: Provided detailed, section-by-section critique highlighting vague statements, misalignments, and gaps in logic, along with suggestions for restructuring and clarification.	Incorporated AI recommendations to revise the problem statement for precision, restructured the research framework to enhance coherence, and clarified methodological descriptions to better align with academic expectations.
ChatGPT	Prompt: "Now, evaluate the Statement of the Problem (SOP) and research questions in the attached document. Identify any ambiguities, inconsistencies, or missing elements, and provide specific,	Used the feedback to revise and sharpen the research questions, included hypotheses where appropriate, and ensured the SOP

	actionable suggestions for improvement and revision. Additionally, assess whether each research question necessitates a hypothesis, offering a clear justification for your determination. Ensure feedback is detailed, constructive, and aligned with rigorous academic standards." Output: Identified vague phrasing, missing elements, and logical gaps. Recommended clearer formulation of research questions and identified which required hypotheses, justifying each with academic rationale.	clearly defined the research gap and objectives.
ChatGPT	Prompt: “Following that, assess the methodology in relation to the SOP. Evaluate whether the proposed methodology effectively addresses the research problem and provide a detailed analysis of its strengths and limitations. Offer specific, actionable suggestions for improvement and revision, ensuring methodological rigor and alignment with research objectives. If potential gaps or inconsistencies exist, suggest alternative approaches or refinements to enhance the study’s validity and reliability.” Output: Provided a breakdown of methodological strengths and limitations, noted gaps in data collection techniques, and suggested improvements to ensure alignment with research goals.	Revised the methodology to address AI-identified gaps, improved clarity in procedure descriptions, and incorporated suggestions to enhance validity, reliability, and alignment with the research objectives.

Dated and Signed:

Estrella, Novelle Lyn / 05-28-25

Signature Above Printed Name/ Date

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

I am Nathaniel L. Magtibay, faculty member/student/administrative employee of the Polytechnic University of the Philippines here declare that I HAVE /HAVE NOT used Artificial Intelligence to complete this research project/thesis/dissertation titled Multi-Head Attention Transformer for Filipino Sign Language.

The use of Artificial Intelligence is in the following manner:

AI Tool Used	Detailed Prompts and AI Outputs	How the output was incorporated in the project
ChatGPT	Prompt: "Summarize this paper." (Prompt was used repeatedly for multiple academic papers related to the research topic.) Output: Generated concise and coherent summaries of academic papers by extracting the key objectives, methodologies, findings, and contributions. These summaries provided a clear understanding of each paper's relevance to the research without needing to read the entire text word for word. In some cases, the tool also highlighted how the papers aligned with or differed from the proposed approach.	Used as starting points to draft the related works and literature review section, helping to compare existing methods with the proposed system. Helped in organizing sources thematically, making it easier to synthesize

Dated and Signed:

Magtibay, Nathaniel L. / 05-28-25

Signature Above Printed Name/ Date

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

I am Jeremias G. Pablo, faculty member/student/administrative employee of the Polytechnic University of the Philippines here declare that I HAVE /HAVE NOT used Artificial Intelligence to complete this research project/thesis/dissertation titled Multi-Head Attention Transformer for Filipino Sign Language.

The use of Artificial Intelligence is in the following manner:

AI Tool Used	Detailed Prompts and AI Outputs	How the output was incorporated in the project
ChatGPT	Prompt: “Should the InceptionV3-GRU model, used only for comparison, be included in the Research Instrument section of our thesis proposal?” Output: Clarified the distinction between the experimental baseline and the actual system. Recommended a brief clarification in the manuscript and suggested a mapping table to align with PUP’s computing problem criteria.	Guided the revision of the Research Instrument section to clarify the role of InceptionV3-GRU. Inspired the development of a mapping table used in defense prep to show compliance with computing-focused research requirements.
ChatGPT	Prompt: "Analyze if our thesis counts as a computing problem based on this definition..." Output: Provided an in-depth evaluation linking the thesis to definitions of computing and computational problems, classifying it under function and optimization problems with academic justification.	The analysis was directly used to revise the "Statement of the Problem" and "Research Design" sections, ensuring alignment with university guidelines that require a computing problem. It also clarified the experimental basis of the comparative model evaluation.

ChatGPT	Prompt: “Evaluate the logical consistency and alignment of the research methodology with the statement of the problem in a thesis proposal comparing MHAM-Transformer and InceptionV3-GRU for Filipino Sign Language Recognition.” Output: Provided detailed critique of methodological gaps, alignment issues, and recommended statistical corrections (e.g., use of paired t-tests, Holm-Bonferroni adjustment).	The analysis directly informed the revision of the Statement of the Problem and Research Design sections, clarifying the computing problem, aligning variables, and justifying the experimental comparison approach, in compliance with academic and institutional research standards.

Dated and Signed:

Pablo, Jeremias G. / 05-28-25

Signature Above Printed Name/ Date

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

I am Rica Joi C. Migueh, faculty member/student/administrative employee of the Polytechnic University of the Philippines here declare that I HAVE /HAVE NOT used Artificial Intelligence to complete this research project/thesis/dissertation titled Multi-Head Attention Transformer for Filipino Sign Language.

The use of Artificial Intelligence is in the following manner:

AI Tool Used	Detailed Prompts and AI Outputs	How the output was incorporated in the project
ChatGPT	Prompt: “Is it acceptable to use both ‘this study aims’ and ‘this study seeks’ in our Purpose Statement?” Output: Identified redundancy and advised choosing one for clarity. Explained tonal and stylistic differences between “aims” and “seeks.”	Helped finalize the thesis Purpose Statement by consistently using “aims,” based on the tone and intent of the paper.
ChatGPT	Prompt: “Can I avoid repeating all statistical procedures in the sections for RQ1 and RQ2 if they are already detailed in the Statistical Treatment section?” Output: Advised to keep full test procedures in a single Statistical Treatment section and refer to it from RQ1 and RQ2 analysis to avoid redundancy.	Structured the thesis such that statistical steps (t-test, Holm correction, Cohen’s d) appear once under “Statistical Treatment,” with concise references in earlier analysis.
ChatGPT	Prompt: “Could you verify whether this Statistical Treatment section aligns with the updated data analysis?” Output: Confirmed overall alignment, suggested clarifying the level of paired difference (sample vs class level), and improving the description of Holm’s correction.	Used to finalize the “Statistical Treatment” section, with clear language about how paired values are prepared and used in the t-test.

Dated and Signed:

Migueh, Rica Joi C. / 05-28-25

Signature Above Printed Name/ Date

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Effective communication is fundamental to human interaction. While spoken language predominates, sign languages provide an essential linguistic channel for Deaf communities worldwide. In the Philippines, Filipino Sign Language (FSL) is the primary visual–gestural medium of the local Deaf population, and its national status was formally recognized through the FSL Act, also known as Republic Act 11106, in 2018 (Notarte-Balanquit, 2021). Yet, in contrast to other national sign languages, FSL remains comparatively under-researched, limiting the development of technologies that might bridge communication gaps between Deaf and hearing Filipinos.

At the linguistic level, FSL conveys meaning through a rich interplay of manual (hand movements) and non-manual (facial expressions, body movements) markers, forming a complete grammar and syntax independent of spoken Filipino (Notarte-Balanquit, 2021; Fischer, 2015; Tiongson & Martinez, 2004). Effective recognition of FSL therefore requires models capable of integrating these spatio-temporal features in both space and time and remaining resilient when some of those signals are temporarily hidden by occlusion—such as when the signer's hands pass in front of the face. Despite the language's critical role in social inclusion, most existing Filipino Sign Language recognition (FSLR) systems still target isolated gestures—typically the manual alphabet or basic vocabulary. As a result, these models struggle to disambiguate signs with similar hand motions but different facial expressions—an essential distinction in natural sign communication—and they degrade sharply whenever keypoints are occluded (Tupal, 2023; Rivera & Ong, 2018).

Conventional deep-learning pipelines based on Convolutional Neural Networks (CNNs) for spatial feature extraction and recurrent units such as Gated Recurrent Units (GRUs) for temporal modelling have achieved steady progress. While this CNN–GRU hybrid approach has shown promise, it processes input sequentially and treats spatial and temporal cues independently. Because these cues are encoded in separate streams, the model cannot fall back on visible features when occlusion masks one modality, leading to frequent misclassifications in realistic, cluttered signing scenarios. This architectural limitation results in poor integration of spatio-temporal features, leading to frequent misclassifications, particularly for signs where facial expression is a critical differentiator (Tupal, 2023; Pigou et al., 2015).

Recent advances in deep learning, especially the Transformer architecture introduced by Vaswani et al. (2017), present a compelling alternative. Unlike recurrent models, Transformers use a Multi-Head Attention Mechanism (MHAM) to process sequences in parallel, allowing the model to attend to multiple aspects of the input simultaneously. This parallelism enables finer-grained integration of spatio-temporal features—such as hand movement trajectories, facial affect, and body posture—offering a richer and more holistic representation of sign language inputs, even when parts of the signer become temporarily invisible (Doshi, 2021; Liu et al., 2023). While Transformers have revolutionised natural-language and vision tasks, their potential has yet to be systematically evaluated for FSL. No existing study has systematically benchmarked MHAM-based Transformers against CNN–GRU architectures in the context of FSL, nor examined their effectiveness in integrating manual and non-manual features for improved sign disambiguation under varying levels of occlusion.

In response, this study addresses that gap by developing a Transformer-based model with MHAM for recognizing isolated Filipino Sign Language glosses and explicitly tests its robustness to keypoint occlusion. The model is evaluated against the

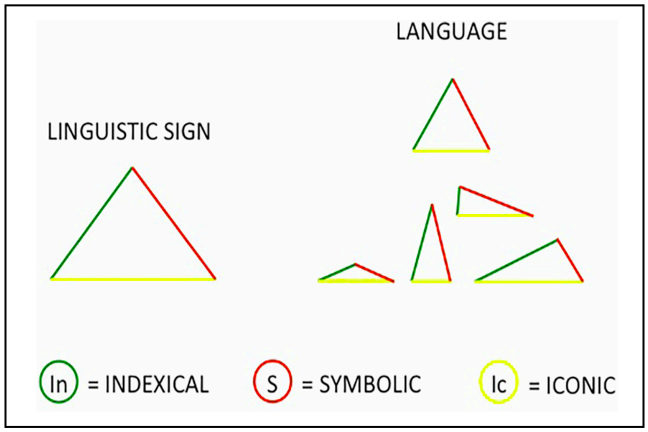
InceptionV3-GRU architecture proposed by Tupal (2023), with particular attention to its ability to integrate spatio-temporal features when some are hidden. By empirically comparing these models across clean and artificially occluded keypoint streams, the research aims to determine whether Transformer-based approaches offer meaningful improvements in recognition and semantic-category classification accuracy, especially for signs that rely heavily on facial expressions and other non-manual cues. Furthermore, by addressing the deficiencies in current evidence and model limitations, this study seeks to contribute to the development of more robust and inclusive FSLR systems, ultimately promoting accessibility and communication equity for the Filipino Deaf community and advancing the application of Transformer models to domain-specific, multimodal tasks.

Theoretical Framework

This research is anchored in deep learning-based approaches for sign language recognition (SLR), with a particular focus on the Transformer architecture introduced by Vaswani et al. (2017). Central to the Transformer is the Multi-Head Attention Mechanism (MHAM), which has demonstrated state-of-the-art performance across a wide range of tasks, including computer vision and the understanding of sign languages.

The Triangular Semiotic Model (Peirce, 1994) posits that meaning is constructed through the relationship between a sign (the form), its object (the referent), and the interpreter (the understanding of the sign). Peirce classifies signs into triadic construction: symbolic, iconic, and indexical categories — each offering a unique way to understand sign language elements. Peirce's model helps explain how technological systems should interpret manual (hand-shape, motion trajectory) and non-manual (facial expressions) features in sign language.

Figure 1. Triangular Semiotic Model – Sign Classification



Symbolic → Arbitrary signs in Sign Language (e.g., fingerspelling or grammatical markers).

Iconic → Gestures that resemble their meaning (e.g., the sign for "eat" mimicking the motion of bringing food to the mouth).

Indexical → Context-dependent signs (e.g., facial expressions modifying the meaning of a sign).

Peircean semiotic features (iconicity, indexicality, and symbolic representation) play a crucial role in determining how linguistic units, including non-manual markers, are used in sign languages (Capirci et al., 2022). This justifies modeling SLR systems to weigh symbolic, iconic, and indexical features dynamically, ensuring accurate interpretation of gestures and their contexts. Integrating Peircean Semiotics into SLR effectively enhances the model to not only detect gestures but also correctly interpret their context and meaning including non-manual markers such as facial expressions, leading to more accurate and context-aware sign recognition systems.

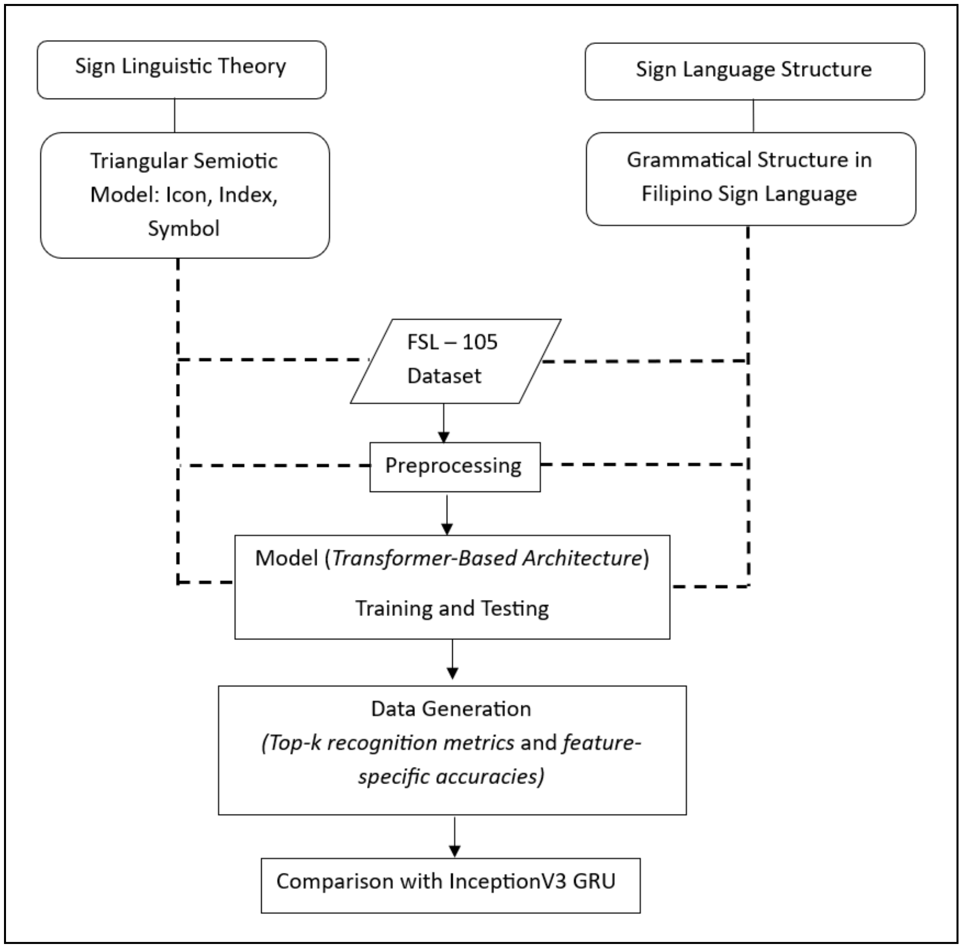
Aligned with this, Filipino Sign Linguistics asserts that FSL is an independent language with its own syntax, grammar, and lexicon, distinct from spoken Filipino and English. It incorporates non-manual cues, such as facial expressions to enhance meaning and context (Notarte-Balanguit, 2021). Incorporating Filipino Sign Linguistics ensures that FSLR systems acknowledge the unique structure and grammar of FSL, distinguishing it from spoken Filipino language.

This study builds upon the recommendations of Tupal (2023), which proposed extending FSL research by utilizing the FSL-105 dataset with Transformer-based models for improved recognition and analysis.

Conceptual Framework

The framework presented in this study builds upon the established theories of the Triangular Semiotic Model and Filipino Sign Linguistics The Triangular Semiotic Model, which emphasizes the symbolic, iconic, and indexical signs, provides a foundation for understanding how visual-spatial cues in FSL carry meaning. This is complemented by Filipino Sign Linguistics, which focuses on the grammatical and spatial structure unique to FSL. These theories guide the recognition and classification of signs in the Filipino context by informing the features extracted during data preprocessing and model training.

Figure 2. Conceptual Model



In this framework, the FSL-105 dataset (Tupal, 2023) is utilized as a source of sign language video samples. Filipino Sign Linguistics provides insight into the unique grammatical rules and spatial structures inherent to FSL, including hand-shape, motion trajectory and facial expressions. Sign Semiotics and Filipino Sign Linguistics guides the recognition and classification of signs by shaping how features are extracted during the data preprocessing phase and how these features are learned and interpreted during model training. Subsequently, a Transformers-based architecture is applied as Tupal (2023) emphasized the importance of evaluating its effectiveness and practicality in real-world FSL scenarios, during model training and testing, chosen for its capacity to capture temporal dependencies and contextual relationships, as theorized in semiotic interpretation.

The theoretical lens of sign semiotics and linguistics is applied throughout, ensuring that model decisions align with how signs are constructed and understood within the FSL framework.

Statement of the Problem

Current FSLR systems, particularly the InceptionV3-GRU remain limited in their ability to robustly model the spatio-temporal features necessary to recognize complex FSL glosses systems treat spatial (hand shape, position) and temporal (motion trajectory) patterns separately and struggle when key visual features are occluded (e.g., hands passing in front of the face), resulting in decreased recognition accuracy under realistic signing conditions.

This study develops a Transformer-based FSLR model equipped with Multi-Head Attention Mechanism (MHAM) to enhance spatio-temporal modeling. Specifically, the study investigates whether such a model can improve performance in both gloss recognition and semantic-category classification compared to the InceptionV3-GRU baseline, under keypoint streams with and without occlusion.

To achieve this, the research is guided by the following central question:

How does a Multi-Head Attention Mechanism Transformer improve recognition and classification of isolated FSL glosses compared to the IV3-GRU baseline, both with and without occlusion?

To explore this question, the study further investigates the following key sub-questions:

1. What is the performance of IV3-GRU and MHAM Transformer in recognizing sign glosses without occlusion in terms of:
    a. Precision
    b. Recall
    c. F_1 Score
2. What is the performance of IV3-GRU and MHAM Transformer in recognizing sign glosses with occlusion in terms of:
    a. Precision
    b. Recall
    c. F_1 Score
3. What is the performance of IV3-GRU and MHAM Transformer in classifying sign glosses without occlusion in terms of:
    a. Precision
    b. Recall
    c. F_1 Score
4. What is the performance of IV3-GRU and MHAM Transformer in classifying sign glosses with occlusion in terms of:
    a. Precision
    b. Recall
    c. F_1 Score

Hypothesis

H_{01}. There is no significant difference in the performance of IV3-GRU and MHAM Transformer in recognizing sign glosses without occlusion in terms of Precision, Recall, and F_1 Score.

H_{02}. There is no significant difference in the performance of IV3-GRU and MHAM Transformer in recognizing sign glosses with occlusion in terms of Precision, Recall, and F_1 Score.

H_{03}. There is no significant difference in the performance of IV3-GRU and MHAM Transformer in classifying sign glosses without occlusion in terms of Precision, Recall, and F_1 Score.

H_{04}. There is no significant difference in the performance of IV3-GRU and MHAM Transformer in classifying sign glosses with occlusion in terms of Precision, Recall, and F_1 Score.

Scope and Limitations of the Study

This study focuses on the development and evaluation of a MHAM-based Transformer architecture for FSLR, with the goal of addressing the limitations observed in the baseline IV3-GRU model. Specifically, the research investigates how effectively each model captures and integrates critical spatio-temporal features such as hand shapes, motion trajectories, and facial expressions—while maintaining recognition fidelity when any of these keypoints become occluded.

The scope of this research is limited to the analysis of video sequences from the publicly available FSL-105 dataset. As such, the study does not involve the creation, collection, or augmentation of new datasets. Additionally, the evaluation will be conducted in an offline experimental setup, and real-time application or deployment is not within the scope of this study. The research is also constrained by available computational resources, which may affect the scalability or depth of model training.

Furthermore, the findings discussed are specifically designed for FSL and may not be directly transferable to other sign languages, which differ in grammar, gestures, and non-manual signals. Finally, the study is centered on technical performance and does not extend to user-centered evaluations or integration with assistive applications in real-world settings.

Significance of the Study

Transformer-based models have shown strong results in sequence modeling tasks, but their use in FSLR has not been explored in depth. FSL is challenging because it relies on hand gestures, facial expressions, and body movement. Current models like IV3-GRU mostly focus on hand movements, often missing these non-manual features. Whether attention-based models like Transformers using MHAM can better handle FSL's complexity is still uncertain. This justifies comparing these approaches to see if newer models can improve recognition accuracy and support real-world FSL use.

The Filipino Deaf community will benefit from this study through the development of more responsive and inclusive communication tools. As accuracy improves, these tools could help Deaf individuals express themselves in everyday scenarios like requesting assistance, using public services, or engaging in classroom activities. By building a foundation for systems that better understand FSL, this study helps bridge the communication gap between Deaf and hearing individuals.

Educators and interpreters will also receive actionable metrics on non-manual cue recognition, identifying which facial expressions, head movements and motion subtleties are most often misclassified, so they can tailor lesson plans, design targeted drills, and build assessment rubrics that directly address these gaps in learner performance.

This research will also benefit developers working on assistive technology. The proposed model is designed to better capture spatio-temporal features needed for accurate sign recognition, making it more suitable for translation apps, learning software, or interpretation tools. With clear benchmarks and evaluation results, the study will provide technical insights developers can apply to improve recognition in future projects.

Lastly, this research will contribute to the ongoing work of future researchers in SLR and machine learning. It will present a comparative evaluation of Transformer-based and IV3-GRU architectures using the FSL-105 dataset. Researchers working on SLR systems can use this study as a foundation for further experimentation and model enhancement.

Definition of Terms

Adam Optimizer. Used in training the models to automatically adjust learning rates and update weights to reduce classification error.

Classification Performance. The model's ability to correctly assign glosses to semantic categories. Evaluated using precision, recall, and F_1-score.

Gloss. A textual label representing the true meaning of a sign in sign language. In this study, each gloss corresponds to a specific FSL (Filipino Sign Language) gesture.

InceptionV3-GRU (IV3-GRU). A baseline model used for comparison, combining InceptionV3 for extracting spatial features from video frames and GRU for analyzing their sequence over time. (Tupal, 2023)

MediaPipe framework. A machine learning toolkit used in this study to track hand positions and body landmarks (MediaPipe Hands and Pose) and perform background removal (MediaPipe Selfie Segmentation) for sign language recognition.

Multi-Head Attention Mechanism (MHAM). A Transformer feature that captures spatial and temporal patterns in sign language videos by attending to hand-shape, motion, and facial cues simultaneously.

Occlusion. The blocking of visual keypoints (e.g., hands, face) where critical facial features like the eyes, eyebrows, nose, and mouth are obscured.

Recognized Sign. Referred to as the predicted or recognized gloss; it is the output label assigned by the model to a given sign input after identifying it.

PyTorch. The machine learning framework used to implement, train, and test all models in this study.

Recognition Performance. The model's ability to correctly identify individual glosses. Evaluated using precision, recall, and F_1-score.

Transformer Architecture. A deep learning model that uses self-attention to process sequences in parallel. In this study, it learns spatio-temporal features in sign language videos, capturing gesture changes over time and space without recurrence.

Vast AI. A cloud computing platform utilized in this study for tool development, including training and experimentation, due to its cost-effectiveness and flexibility in offering a wide range of GPU options.

Weights and Biases (W&B). Used to log training metrics, visualize model performance, and manage experiments throughout the study.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

Recent advancements in SLR have introduced various methods to improve communication accessibility for the Deaf community, especially in recognizing complex languages like FSL. This section highlights major topics including general SLR, FSL-specific challenges, existing approaches in SLR, current issues in model performance, and the use of both CNN-RNN hybrids like IV3-GRU and emerging Transformer-based models for improved recognition accuracy. With these developments in mind, the study centers on applying these approaches to the development of more robust and inclusive FSLR systems.

Sign Language Recognition

Sign language is a complex visual form of communication that utilizes manual gestures, facial expressions, and body movements to convey meaning (Ugale et al., 2023; Batnasan, 2024). It serves as the principal form of communication for those who are deaf or hard of hearing. The historical context reveals a growing advocacy for the recognition of signed languages as legitimate linguistic systems (Batnasan, 2024). Automated SLR is a multidisciplinary field at the intersection of computer vision, machine learning, and natural language processing (Batnasan, 2024; Liu et al., 2023; Pigou et al., 2015). Research on SLR creates systems that can automatically decipher and translate sign language into spoken or written language, enabling smooth communication.

Major milestones in SLR include the early adoption of Hidden Markov Models (HMMs) inspired by automatic speech recognition (De Coster et al., 2020; Pigou et al., 2015). However, in recent years, there has been a considerable movement towards deep learning algorithms that have shown greater performance in related areas such as image classification and speech recognition. Primary terms in SLR frequently differentiate between isolated SLR, which focuses on identifying individual signals, and continuous SLR (CSLR), which scores to detect sequences of signs in a continuous stream (De Coster et al., 2020; Batnasan, 2024). The development of accurate and resilient automatic recognition systems is extremely important since it promises to improve accessibility, accelerate the annotation of sign language corpora, and simplify communication in a variety of real-world applications.

Occlusion.

The head serves as a crucial place of articulation in sign languages, and identifying when and where the hand occludes the head can provide valuable insights into the intended sign structure. An occlusion begins when critical facial features, such as eyes, eyebrows, nose, or mouth, are hidden and ends once these features become visible again.

However, not all occlusion events are intentional or linguistically meaningful. Some occur incidentally, when the hand passes in front of the face without contact (incidental non-contact occlusion), while others involve partial or complete facial coverage, where the hand targets one area, like the nose, but also covers adjacent regions, such as the mouth or neck (overlapping target-contact occlusion). A third category, full-face occlusion, describes cases in which the hand or forearm simultaneously obscures the eyes, nose, and mouth, temporarily masking the signer’s facial expression altogether. Although human observers can easily interpret such variations, automated systems often struggle to accurately recognize signs when occlusions are misclassified or missed altogether (Viitaniemi et al., 2013).

Occlusion also poses significant challenges in facial feature tracking for sign language recognition, especially when facial regions are temporarily blocked by hands or other objects like hair. Many signs require the hands to move rapidly in front of the face, which complicates automated analysis. Unlike humans, who can fluidly interpret these fast movements, computer systems rely on discrete video frames, making it difficult to reconstruct hand trajectories during brief occlusions (Castelli et al., 2005).

Confidence Level.

In machine learning, particularly in classification tasks such as Sign Language Recognition (SLR), the confidence level refers to the probability score that a model assigns to its prediction, indicating how certain it is about the result. On the other hand, the threshold is a predefined value used to decide whether to accept or reject a model's prediction based on its confidence. If the model's confidence score meets or exceeds the threshold, the prediction is accepted; otherwise, it may be considered uncertain or discarded.

The optimal threshold value is not fixed and often varies depending on the specific application and context. Factors such as the acceptable risk level, user expectations, and operational constraints all influence the ideal threshold setting.

According to Wenkel et al. (2021), selecting appropriate thresholds tailored to the deployment scenario is important, especially in safety-sensitive applications like robotics, medical, or edge devices. Proper threshold selection is critical for applications where false positives and negatives have severe consequences. Relying solely on standard metrics without adjusting the confidence threshold might lead to suboptimal or unsafe deployment decisions. Thus, certain approaches involve utilizing the Probability-based Detection Quality metric to identify an optimal confidence score threshold that balances true positives, false positives, and false negatives, analyzing performance across various thresholds to determine the most favorable point, and considering application-specific priorities to select a threshold that ensures fair model comparison and effective deployment. Such approach aids in determining the most suitable confidence level for fair model comparison and deployment in the real-world.

Filipino Sign Language Recognition

In accordance with R.A. 11106, also known as the FSL Act of 2018, FSL is officially recognized as the Philippines' national visual-gestural language. Consequently, scholarly conversations on deafness have broadened, and FSL itself has gained opportunities to advance and achieve recognition as a language system fundamental to the formation of the Filipino Deaf identity. Alongside its status as a natural language of the Deaf (Notarte-Balanquit, 2021), FSL continues to establish its significance and prominence.

Despite this progress, research on FSLR remains in its early stages compared to more extensively studied sign languages such as American Sign Language and Indian Sign Language (Tupal, 2023). Recent notable developments in the local context include the creation of larger FSL video datasets and the exploration of deep learning models to improve recognition accuracy (Tupal, 2023; Batnasan, 2024). However, fully capturing the complexity of FSL—particularly non-manual signals and individual variations in signing—remains a challenge (Tupal, 2023; Rivera & Ong, 2018).

Existing Approaches in Sign Language Recognition

SLR has progressed significantly, as numerous approaches have been explored to bridge the communication gap between the Deaf and hearing communities.

Researchers have investigated sensor-based methods, Kinect-based systems, vision-based approaches, and hybrid techniques, each contributing unique advantages yet facing specific limitations.

Sensor-Based Methods.

Sensor-based methods, such as data gloves and wearable devices, have demonstrated high accuracy—reaching up to 96% (Oliva et al., 2018)—through capturing the intricate dynamics of sign language (Abhishek et al., 2016; Mehdi & Khan, 2002). However, they often lack practicality due to their bulkiness and expense. Moreover, because these systems focus primarily on hand movements, they neglect non-manual cues such as facial expressions and body language, which are integral to sign language (Oliva et al., 2018).

Kinect-based systems utilize depth-sensing technology to achieve similarly high accuracy, at about 95%, for capturing 3D gesture data (Oliva et al., 2018). Despite this, they tend to overlook fine details in hand movements and facial expressions, reducing their effectiveness in fully recognizing the complexities of sign language.

Vision-Based Approaches.

Vision-based techniques offer a more practical alternative. Dynamic Time Warping (DTW) and Support Vector Machines (SVM) have been employed alongside Kinect for smaller sign vocabularies (Oliva et al., 2018). While SVM has been found more efficient than DTW, scaling these methods to larger vocabularies remains challenging. Manifold projection techniques have demonstrated promising results, achieving high recognition rates on small sets of signs (Cabalfin et al., 2012), yet their accuracy drops significantly with larger vocabularies. CNNs have also gained traction, successfully recognizing digits and limited vocabularies (Jarabese et al., 2021;

Montefalcon et al., 2021). Nonetheless, a recurring limitation is the focus on a small number of sign classes and, in earlier studies, a reliance on static images that fail to capture sign language's dynamic nature. These models also tend to emphasize hand gestures, frequently overlooking non-manual signals (Oliva et al., 2018).

Hybrid Approaches.

Hybrid methods combine sensor-based and vision-based techniques to capture a more comprehensive range of both manual and non-manual cues. Some studies have employed Artificial Neural Networks (ANN) and SVM to include non-manual signals for FSLR (Rivera & Ong, 2018). However, these approaches generally have a limited scope, and achieving higher accuracy in recognizing the intended meaning or gloss remains problematic.

Challenges and Issues in Sign Language Recognition

Al-Qurishi et al. (2021) emphasize that despite the progress of these approaches, major challenges remain, notably the scarcity of high-quality, large-scale datasets—especially for less-studied sign languages like FSL. While this issue is undoubtedly significant, addressing it falls outside the scope of the present study. Many models do not adequately address non-manual signals, and much of the research focuses on isolated sign recognition rather than continuous signing. In addition, practical applications and thorough evaluations in diverse, real-world environments are lacking. Other challenges include individual and regional variability within sign languages, the visual similarity of certain signs, constraints on real-time processing, and the difficulty of adapting models across different sign languages (Oliva et al., 2018).

InceptionV3-GRU: CNN-RNN Hybrid Models for SLR

CNNs and Recurrent Neural Networks (RNNs) have been widely used in SLR due to their complementary strengths in feature extraction and sequence learning. CNNs excel at extracting spatial features from images by detecting edges, shapes, and patterns through convolutional, pooling, and fully connected layers. As noted by Ugale et al. (2023), CNNs reduce the need for manual preprocessing by automatically identifying relevant image features, making them particularly effective for static hand gesture recognition.

Meanwhile, RNNs, especially Long Short-Term Memory (LSTM) networks, are well-suited for processing sequential data by maintaining dependencies across time steps, making them ideal for recognizing continuous sign language gestures. Lipton et al. (2015) emphasize that LSTMs mitigate issues such as vanishing and exploding gradients, ensuring better performance on larger datasets.

The integration of CNNs and RNNs in SLR typically follows a two-step process: CNNs first extract spatial features from sign language video frames, and then RNNs model the temporal dependencies of gestures for accurate sequence prediction. CNNs-RNNs approach has been widely applied in various gesture and action recognition tasks. Ugale et al. (2023) discuss how CNNs handle static hand gestures, while RNNs, particularly LSTMs, enhance recognition by processing temporal sequences. Furthermore, hierarchical attention networks and sequence-to-sequence RNN models have been employed to translate signing videos into text by analyzing sequential frames.

Lipton et al. (2015) also highlight the application of CNN-RNN models in general action recognition, where movements are tracked across multiple frames to enable real-time gesture interpretation. With the strengths of both architectures, CNN-RNN models significantly enhance recognition accuracy and efficiency in sign language applications.

InceptionV3-GRU Model in SLR.

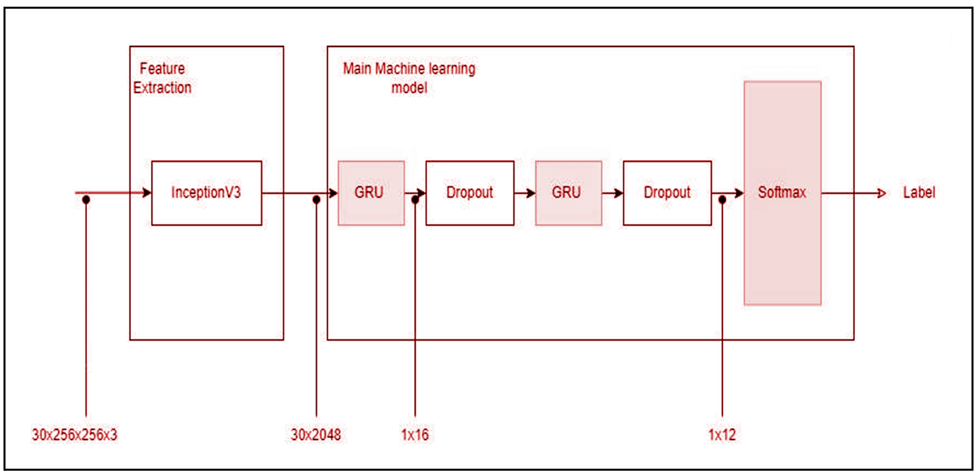
The IV3-GRU model stands as one of the three validated architectures developed for SLR in the study conducted by (Tupal, 2023), combining the sophisticated spatial feature extraction capabilities of CNNs with the temporal sequence modeling strengths of GRUs. The model utilizes IV3, a state-of-the-art CNN architecture designed by Google to efficiently capture multi-scale features while drastically reducing the number of parameters compared to earlier networks such as AlexNet. The adoption of IV3 is motivated by its proven ability to balance accuracy and computational efficiency—critical for handling video frames in sign language gestures.

The input video undergoes preprocessing where each frame is resized to 256x256 pixels with the background removed using MediaPipe Selfie Segmentation, effectively isolating the signer. From each video, 30 evenly spaced frames are sampled at a rate of 7 frames per second. Each frame is then fed through the IV3 model, which converts it into a 2048-dimensional feature vector representing high-level spatial characteristics. Concatenating these vectors creates a 30x2048 input matrix that encapsulates the spatial information across the temporal length of the video clip.

To capture the temporal dynamics inherent in sign language gestures, the sequence of feature vectors is inputted into a two-layer GRU network. The GRU layers, containing 16 and 12 hidden units respectively, offer an effective solution to the vanishing gradient problem common in recurrent architectures, enabling the model to learn long-range dependencies within the video sequence. Furthermore, compared to LSTMs, GRUs require fewer parameters and converge faster during training, making them especially suitable for SLR.

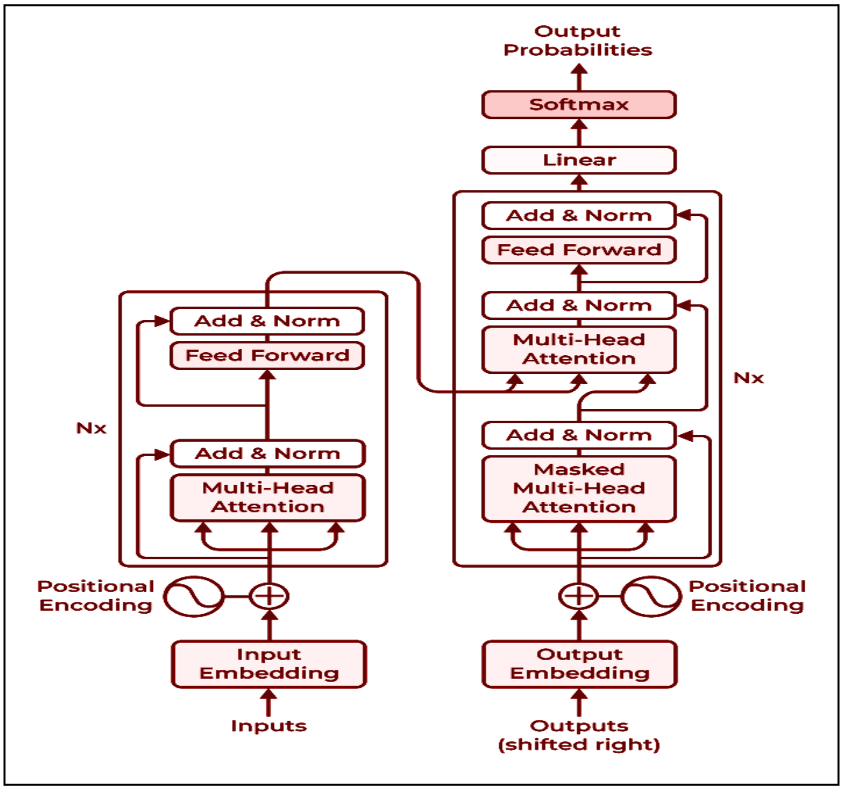
Dropout regularization is incorporated after each GRU layer with a dropout rate of 0.3 to reduce overfitting. The sequential features are finally passed through a softmax classification layer that outputs the probabilities across the sign classes. The hybrid architecture—IV3 for spatial feature extraction and GRU for temporal modeling—effectively bridges spatial and temporal modalities, resulting in a model for FSLR with higher accuracy and a reduced parameter count compared to prior works (Tupal, 2023).

Figure 3. InceptionV3-GRU Architecture



Transformer-based Approaches for Sign Language Recognition

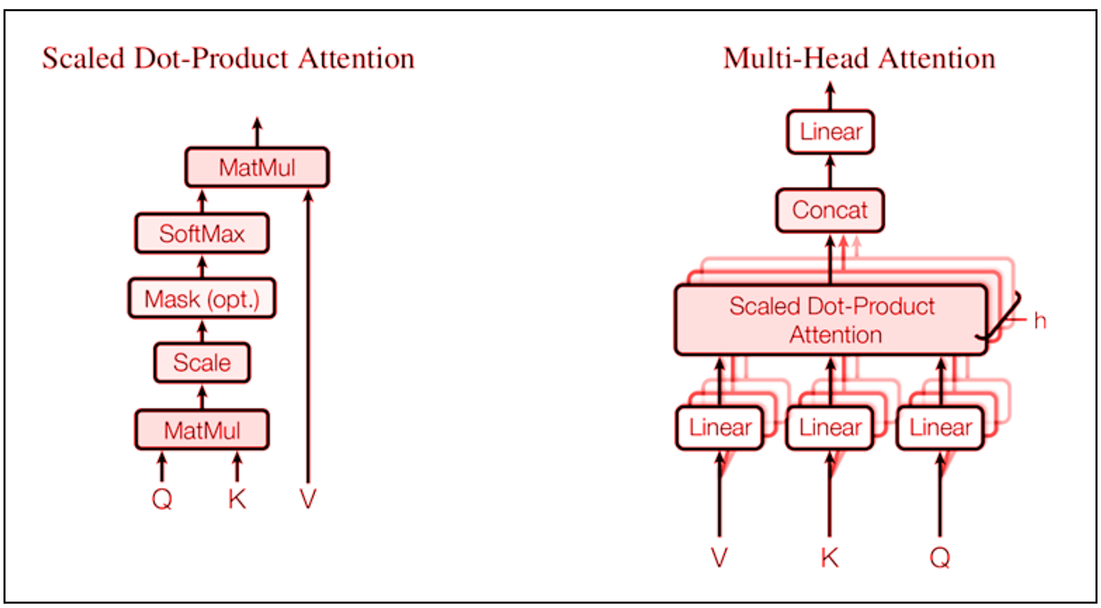
Figure 4. Transformer Encoder-Decoder Architecture



According to Vaswani et al. (2017), the Transformer model revolutionized deep learning for sequence-based tasks by relying entirely on self-attention mechanisms, eliminating the need for recurrent or convolutional layers. Unlike traditional RNNs and LSTM networks, which process data sequentially, Transformers enable parallel processing, significantly improving computational efficiency and scalability. At the core of the Transformer architecture is the self-attention mechanism, which allows the model to weigh the importance of different elements in a sequence regardless of their position. The self-attention mechanism enables the model to capture long-range dependencies more effectively than RNNs, where information may degrade over long sequences due to vanishing gradients. The standard Transformer consists of an encoder-decoder architecture: The encoder processes the input sequence into a continuous representation, utilizing stacked self-attention layers and feed-forward neural networks. The decoder generates the output sequence by attending to both the encoder's outputs and previously generated tokens, ensuring accurate sequence prediction. A key feature of Transformers is multi-head attention, which allows the model to capture diverse contextual relationships by attending to different parts of the sequence simultaneously. Additionally, positional encoding is used to retain sequence order information, as the model lacks inherent recurrence.

The self-attention mechanism stands as a cornerstone of the transformer architecture, facilitating each element in the sequence to attend to every other element, thereby capturing intricate relationships and dependencies irrespective of their relative positions (Batnasan, 2024).

Figure 5. Scaled Dot-Product Attention & Multi-Head Attention



Scaled Dot-Product Attention, introduced by Vaswani et al. (2017), is a core component of the Transformer model. It operates on three primary inputs: queries (Q), keys (K), and values (V), which are typically represented as matrices when processed in batches. The attention mechanism computes the dot product between each query and all keys to measure similarity, scales the result by the square root of the key dimension ($\sqrt{dk}$), and applies a softmax function to produce attention weights. These weights are then used to compute a weighted sum of the values, generating the attention output. The scaling factor is crucial, as it prevents large dot product values—which can occur with high-dimensional keys—from pushing the softmax function into regions with vanishing gradients, thus stabilizing training and improving learning efficiency.

To handle multiple queries simultaneously, the model performs these computations using efficient matrix operations. Building on Scales Dot Product Attention, Vaswani et al. (2017) also introduced Multi-Head Attention, which enables the model to capture a broader range of relationships within the input sequence. Instead of performing a single attention operation, Q, K, and V are projected into multiple lower-dimensional subspaces, each processed by an independent attention head. These heads compute attention in parallel, allowing the model to focus on different aspects of the input simultaneously. The outputs from all heads are concatenated and passed through a final linear transformation, preserving the original dimensionality while enriching the representation.

In the original Transformer architecture, eight attention heads are used, with each head having a dimension of 64 ($dk = dv = dmodel/8$). This design ensures that the computational cost remains comparable to single-head attention while significantly increasing the model's capacity to learn complex patterns. As Batnasan (2024) highlights, the mechanism of Transformer Architecture allows the model to attend to different positions within the input sequence concurrently, capturing diverse contextual relationships and improving recognition tasks such as sign language understanding.

Since the Transformer lacks recurrence and convolution, it requires a way to incorporate the order of the sequence. This is achieved through positional encoding, which is added to the input embeddings. The positional encodings are computed using sine and cosine functions of different frequencies, allowing the model to learn relative positions in the sequence. Positional Encoding ensures that the model can differentiate between different positions even when processing the entire sequence simultaneously (Vaswani et al., 2017).

According to Batnasan (2024), transformers offer several advantages over CNN-RNN architectures, particularly in SLR. One key advantage is their ability to capture both short- and long-range dependencies across an entire sequence using self-attention mechanisms. Unlike RNNs, which process sequences step-by-step, transformers analyze all frames simultaneously, leading to improved context awareness. Additionally, transformers enable parallel processing, whereas CNNs and RNNs process data sequentially or locally, which limits computational efficiency. The parallelism significantly improves processing speed and makes transformers more suitable for real-time applications. Another strength of transformers is their ability to leverage pretraining on large datasets, which enhances their generalization capabilities. In contrast, CNNs and RNNs often require domain-specific training data, making them less adaptable to new datasets. Transformers also naturally accommodate sequences of varying lengths, unlike RNNs, which struggle with long sequences and require truncation or padding. Furthermore, transformers achieve competitive performance with fewer parameters compared to 3D CNNs, which require extensive computations due to their spatiotemporal convolutions. This makes transformers more scalable and computationally efficient. They also demonstrate greater robustness to input variability, such as different signing speeds, whereas CNNs and RNNs require extensive data augmentation to handle such differences. Transformers offer improved interpretability through their attention mechanisms, allowing for visualization of influential parts of the input, whereas CNNs and RNNs are more challenging to analyze due to convolutional filters and vanishing gradient issues. These advantages make transformers particularly well-suited for SLR tasks, where capturing temporal dependencies, enabling parallel computation, and handling variable input lengths are crucial (Batnasan, 2024).

Multi-Head Attention in SLR.

Multi-head attention, introduced by Vaswani et al. (2017), extends the self-attention mechanism by performing multiple attention operations in parallel—each referred to as a "head." Each head independently learns to focus on different parts or aspects of the input by using separate learned linear projections of the queries, keys, and values. These diverse attention heads allow the model to capture a wide range of relationships within the input sequence. The outputs from all heads are then concatenated and passed through a final linear transformation, enabling the model to jointly attend to different representation subspaces. This architecture significantly enhances the model's ability to capture complex dependencies in the data, without substantially increasing computational cost. As a result, multi-head attention is a core innovation that underpins the effectiveness of Transformer models in sequence-based tasks.

Attention mechanisms, in general, dynamically highlight the most relevant parts of an input sequence, making them particularly effective for capturing spatial and temporal dependencies. In the context of SLR, this ability is crucial for understanding the nuanced movements of hands, facial expressions, and body gestures over time (Batnasan, 2024). Traditional models often relied on manually crafted features to extract important visual cues from video frames. In contrast, Transformer-based models can automatically learn and focus on these features, greatly reducing the need for manual feature engineering (Vaswani et al., 2017). Vision Transformers (ViTs), in particular, apply self-attention mechanisms to image patches by treating them as tokens in a sequence, similar to words in natural language processing. This design enables ViTs to excel at tasks like object recognition and scene understanding, making them especially well-suited for SLR. By capturing both spatial and temporal information, these models are capable of recognizing continuous sign language expressions with high accuracy. They achieve this by attending to key regions of a signer's body and tracking movement patterns across frames (Batnasan, 2024).

The MHAM plays a critical role in this process by allowing the Transformer to analyze different positions and features concurrently. Each head performs self-attention independently, and their outputs are concatenated to preserve dimensional consistency. This approach, using scaled dot-product attention with keys (K), queries (Q), and values (V), enables the model to focus on the most informative aspects of the sequence and understand contextual relationships more effectively (Vaswani et al., 2017).

In some advanced systems, combining Transformers with convolutional or recurrent neural network architectures has further improved performance. These hybrid models leverage the spatial feature extraction capabilities of CNNs and the temporal modeling strengths of Transformers. Such integrations result in more robust SLR systems, capable of handling complex visual and temporal variations across diverse datasets (Vaswani et al., 2017).

Review of Transformer-Based SLR Studies.

Recent studies on Transformer-based SLR models have demonstrated significant improvements in capturing both spatial and temporal dependencies, leading to more accurate and efficient recognition systems.

Table 1

Comparison of Transformer, CNN, and RNN Models on Diverse Sign Language Datasets

Study	Language / Dataset	Model(s) Compared	Transformer Type / Approach	Results / Accuracy	Key Insights
Batnasan (2024)	WLASL2000, Revised WLASL, ASL-Citizen, Arabic Sign Language Dataset	ViT-B 16 vs ViT-B 32 vs S3D vs VK-Net vs X-CLIP vs Other CNNs	Vision Transformer (ViT), Pretrained on large datasets, Region/extraction modules, Data Augmentation	61.46% on WLASL2000 (proposed), 81.26% on revised WLASL outperforming several baseline models	Incorporating region extraction (head and hands), large-scale pre-training, supervised fine-tuning enhances accuracy
De Coster et al. (2020)	Flemish Sign Language (VGT) / 18730 samples, 100 classes	LSTM baseline, PTN, VTN, MTN	Transformer-based architectures (including Multi-head Attention) and CNN hybrids	Best: 74.7% on 100 classes	Transformers outperform LSTMs in SLR; spatio-temporal features attention enhances accuracy; larger datasets improve performance; end-to-end learning extracts more salient features than keypoints alone
Heuju & Rijal (2022)	Nepali Sign Language (NSL)	CNN-RNN (DenseNet-121 + LSTM) vs. Transformer	Transformer model using DenseNet-121 features	Best accuracy with Transformer: **89.14%** (with 9 signs, 20 frames)	Transformers more effectively capture spatial and temporal dependencies in NSL recognition tasks
Shin et al. (2023)	Korean Sign Language (KSL)	Deep CNNs vs. Transformer-CNN hybrid	Hybrid model with multi-branch attention combining CNNs (local features) and Transformers (global context)	**Transformer-CNN hybrid : 89.00%** (benchmark) **98.30%** (lab dataset)	Hybrid model significantly outperforms previous CNN-based KSL models (79.80%–83.66%)
Kothadiya et al. (2023)	Indian Sign Language (ISL)	CNN and hybrid models vs. SIGNFORMER	SIGN FORMER: ViT-based model with patch embeddings, MHSA, MLP, and strong data augmentation (no convolutions used)	SIGN FORMER: **99.29%** in just 5 epochs	SIGNFORMER achieves high accuracy with fewer epochs, improved efficiency, and robustness to variations; superior to CNN and hybrid models

Synthesis of the Reviewed Literature and Studies

Transformer architectures have demonstrated significant success in natural language processing and computer vision; however, their application in FSLR remains underexplored. A study by Boháček and Hrúz (2022) applied pose-based Transformer models to word-level SLR, achieving a top-1 accuracy of 63.18% on WLASL. Despite these advancements, research on Transformer-based approaches in FSLR is minimal, with most existing models relying on CNNs and RNNs. Expanding Transformer applications in this field could significantly improve recognition accuracy, particularly in capturing sequential dependencies and spatial relationships in signing. As Tupal (2023) suggests, implementing Transformer-based models using the FSL-105 dataset could help extend research in the FSL domain.

According to Batnasan (2024), SLR has faced numerous complex challenges. Chief among these is the inherent complexity of sign languages themselves, which encompass not only intricate hand gestures but also nuanced facial expressions and dynamic body movements. Thus, accurately recognizing and interpreting these spatio-temporal features remains a significant hurdle.

CNN-RNN hybrids have demonstrated strong performance in SLR tasks, such as the IV3-GRU model, which outperformed other architectures in recognizing FSL gestures (Tupal, 2023). Most FSLR models currently employ CNNs, RNNs, or hybrid architectures, as the application of Transformer-based models remain unexplored. In addition, no direct comparison exists between these hybrid models and Transformer-based architectures, leaving a significant gap in understanding their relative effectiveness. Furthermore, the lack of benchmarking makes it difficult to accurately assess their comparative performance.

Chapter 3

METHODOLOGY

Research Design

This study uses a quantitative experimental framework to compare two model architectures—our Transformer with Multi-Head Attention versus the IV3-GRU baseline—on the same FSL-105 dataset under identical preprocessing and training conditions.

The independent variable is the model architecture (Transformer vs. IV3-GRU). The dependent variables correspond directly to our research questions:

1. **Recognition performance (RQ1, RQ2):** Measured by macro-averaged precision, recall, and F_1-score across all 105 gloss classes, evaluated under two conditions:
    a. RQ1: Non-occluded signs
    b. RQ2: Occluded signs
2. **Classification performance (RQ3, RQ4):** Measured by macro-averaged precision, recall, and F_1-score across semantic categories (e.g., greetings, numbers), also evaluated under:
    a. RQ3: Non-occluded signs
    b. RQ4: Occluded signs

Both models are trained from scratch on the same 80/20 training/testing split, using identical hyperparameters and preprocessing steps—including 78-keypoint extraction via MediaPipe, coordinate normalization, and embedding generation.

During evaluation, the held-out test set is automatically divided into non-occluded and occluded subsets based on keypoint visibility (e.g., missing facial landmarks such as the eyes, nose, or mouth). Predictions for recognition and classification tasks are logged and compared against ground truth.

Statistical significance of performance differences between the two models is tested using paired t-tests or Wilcoxon signed-rank tests ($\alpha = 0.05$), depending on data normality.

Sources of Data

This study will utilize the FSL-105 dataset, a publicly available and pre-annotated video dataset specifically developed for FSLR. The use of this existing dataset provides a standardized benchmark for evaluating the performance of both the Transformer-based FSLR model and the baseline IV3-GRU. By adopting the same dataset, this research ensures consistency in experimental conditions and enables a fair, reliable comparison of the Transformer-based model's performance in FSLR.

Dataset Composition.

The FSL-105 dataset consists of 2,130 video samples representing 105 distinct FSL signs. These signs were selected through purposive sampling, a non-probability sampling technique chosen to ensure the inclusion of signs that are pedagogically relevant and frequently encountered by FSL learners. This method was employed to deliberately target core vocabulary essential for functional daily communication, rather than relying on random or exhaustive sign selection (Tupal, 2023). The signs are grouped into ten thematic categories such as Greetings, Survival, Numbers, Calendar, and Family, reflecting practical and contextually meaningful language usage. A selection of representative signs is shown in Table 2.

Table 2

FSL-105 Dataset Composition

Category	Signs (Examples)
Greeting	Good morning, Good evening, How are you, Nice to meet you
Survival	I understand, I know, Yes, No, Wrong, Correct, Slow, Fast
Number	One, Two, Three, Four, Five, Six, Seven, Eight, Nine, Ten
Calendar	January, February, March, April, May, June, July, August, September
Days	Monday, Tuesday, Wednesday, Today, Tomorrow, Yesterday
Family	Father, Daughter, Grandfather, Auntie, Cousin, Parents
Relationships	Boy, Girl, Woman, Deaf, Hard of Hearing, Blind, Married
Color	Blue, Green, Red, Brown, Black, White, Yellow, Orange
Food	Bread, Egg, Fish, Meat, Chicken, Spaghetti, Rice, Longanisa
Drink	Hot, Cold, Juice, Milk, Coffee, Beer, Wine, Sugar, No sugar

The dataset captures both manual features (hand-shape, motion, positioning) and non-manual features (facial expressions, head movements, body posture), which are crucial for accurately interpreting FSL signs. Non-manual features are particularly important in FSL, as they add contextual meaning that can alter the interpretation of a sign.

Participants in the Dataset.

The dataset was recorded with four native FSL signers—two males (one of whom is an FSL expert) and two females. The participants are Deaf and all are fluent in FSL. The decision to limit participation to four individuals was based on methodological and cultural considerations. As the Filipino Deaf community is part of a vulnerable sector, the selection process prioritized ethical engagement, linguistic authenticity, and cultural accuracy over sample size. This is consistent with ethical research practices that emphasize community-informed participation, cultural sensitivity, and the importance of building trust within marginalized groups (Singleton et al., 2014).

The FSL expert, Sir Rey Alfred Lee, selected three of the participants and ensured that all recorded signs reflected real-world, community-accepted usage. A small, curated group also minimized variability in signing styles—reducing ambiguity and enhancing data consistency for recognition and modeling. This controlled approach supports both ethical and methodological integrity, and aligns with established practices in sign language corpora, where expert-guided curation and signer consistency take precedence over larger, less controlled samples.

Prior to recording, all participants were fully informed of the dataset's purpose and provided written consent. A trusted FSL interpreter facilitated the consent process to ensure clarity and transparency.

Dataset Collection and Preprocessing.

The dataset was originally collected under controlled conditions, following the strict protocols established in the foundational study by Tupal (2023), which this research builds upon.

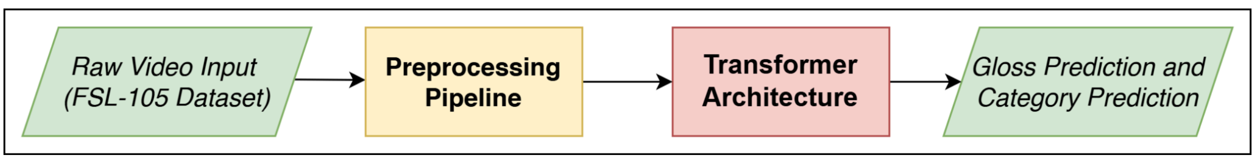
Recordings were conducted in a standardized environment with a plain background to reduce distractions. The signers' torso and hands remained fully visible to capture both manual and non-manual cues. A front-facing camera ensured consistent framing across all videos.

Each sample is a four-second video clip recorded at 60 FPS and 1920×1080 resolution using an iPhone SE 2020. In accordance with Tupal (2023), the dataset was split into 80% training data (1,704 videos) and 20% testing data (426 videos). For efficiency, videos were resized to 640×360 pixels and downsampled to 30 FPS. Keypoint extraction was performed using MediaPipe Pose Estimator, producing 78 keypoints per frame to track key body landmarks. MediaPipe Selfie Segmentation was applied to isolate the signer from the background and emphasize relevant visual cues. Each frame was then padded to a square aspect ratio and resized to 256×256 pixels for consistency across the dataset.

Research Instrument

System Architecture.

Figure 6. System Architecture



The system begins by sampling each video clip from FSL-105 at 30 fps and applying background removal via MediaPipe Selfie Segmentation. For every frame, MediaPipe Pose and Hands modules extract 78 two-dimensional keypoints (25 upper-body, 21 per hand, 11 face) to capture manual and non-manual cues. These (x,y) coordinates are normalized to [–1, 1] and concatenated into a 156-element vector per frame.

Because our input is a structured stream of keypoints rather than raw image patches, off-the-shelf Vision Transformer checkpoints (e.g., ImageNet-pretrained ViT) are not directly applicable and lack the specialized spatio-temporal attention we require. We therefore implement and train a bespoke Transformer that operates on keypoint embeddings from scratch (Dosovitskiy et al., 2020).

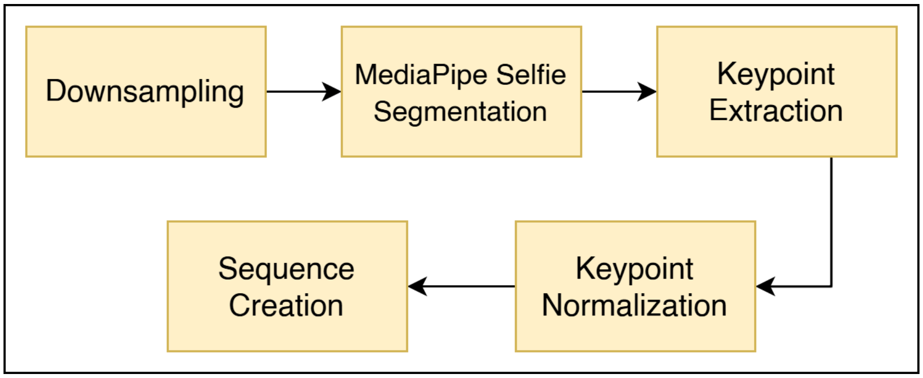
Each frame vector is passed through a learnable linear layer to produce a 768-dimensional embedding. A matching 768-dimensional sinusoidal positional encoding is added to each frame embedding, enabling the model to distinguish frame order without recurrence. The resulting sequence of T embeddings is fed into a stack of six Transformer encoder blocks. Each block applies multi-head self-attention with eight heads (head dim = 64), followed by a position-wise feed-forward network of inner dimension 3072. Layer normalization, residual connections, and dropout ($p = 0.1$) surround both sublayers, stabilizing training

After the sixth block, the sequence is reduced via a learned "CLS" token pooling to a single 768-dimensional vector that fuses all manual and non-manual cues. This fused representation feeds two parallel, lightweight heads. The recognition head applies a linear projection ($768 \rightarrow 105$) and softmax to yield per-gloss probabilities. The category head applies a separate linear projection ($768 \rightarrow C$) and softmax to output probabilities over semantic categories (e.g., greetings, numbers).

By using self-attention over all frames and modalities, the model directly captures synchronous interactions even when some keypoints are missing due to occlusion. Limiting the design to a single pooled output and two simple heads keeps the parameter count moderate (≈ 90 M) and ensures training focuses squarely on spatio-temporal features for gloss recognition and category classification.

Preprocessing Pipeline.

Figure 7. Preprocessing Pipeline

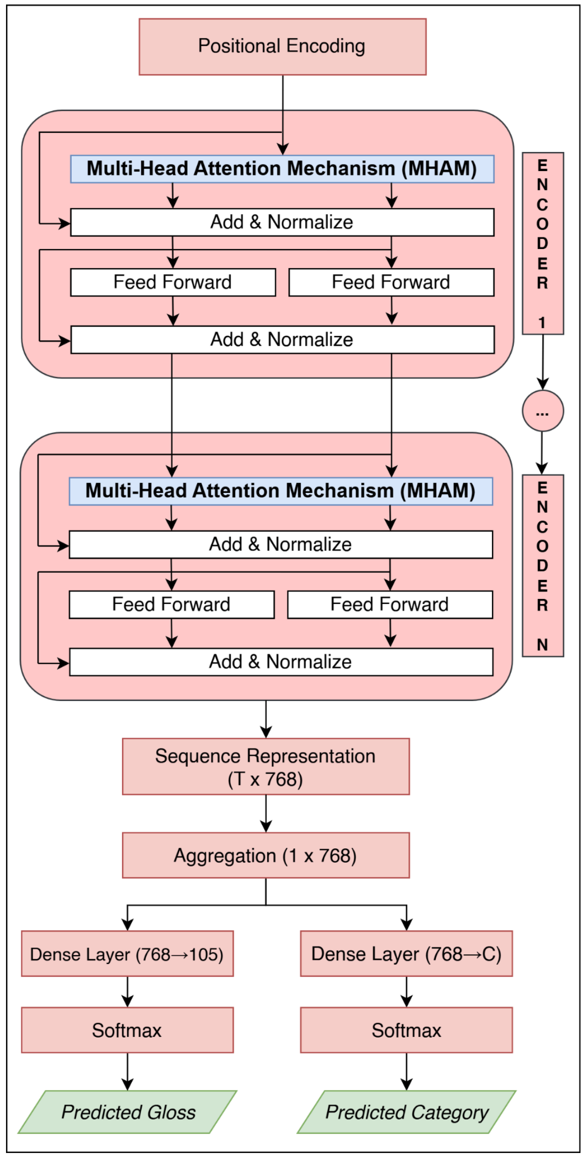


Each 4-second video clip is sampled at 30 fps and resized for consistency. MediaPipe Selfie Segmentation removes the background to isolate the signer. For each frame, MediaPipe Pose and Hands modules extract 78 two-dimensional keypoints (25 upper-body, 21 per hand, 11 facial) to capture manual and non-manual cues. This step captures both manual and non-manual cues while discarding irrelevant features such as background and skin tone (Co et al., 2024).

The raw (x,y) coordinates are normalized—centering and scaling them to a common range. The resulting 156-element vectors per frame are then organized into a time-ordered sequence and passed through a learnable linear layer to project them into the model's embedding space.

Transformer Architecture.

Figure 8. Transformer Architecture



The Transformer processes the frame-level embeddings produced by the preprocessing pipeline. Before entering the encoder, each 768-dimensional embedding is combined with a matching 768-dimensional sinusoidal positional encoding to inject information about its place in the sequence. This augmented sequence of embeddings is then fed into a stack of six encoder blocks, each consisting of multi-head self-attention, a position-wise feed-forward network, residual connections, layer normalization, and dropout.

These sinusoidal encodings generate a unique pattern of sine and cosine values at varying frequencies for each frame index, allowing the model to distinguish both absolute positions (e.g. frame 1 vs. frame 30) and relative distances (e.g. two frames three steps apart). When added to the frame embeddings, they enable the self-attention layers to reason about the order and spacing of frames, essential for capturing the timing and flow of a sign gesture, even though the core Transformer has no recurrent or convolutional structure.

The model consists of an encoder composed of stacked layers, each containing a MHAM, a position-wise feed-forward network, residual (skip) connections, and layer normalization to stabilize training (Zhang & Jiang, 2024). Within each encoder layer, MHAM enables the model to attend to multiple positions in the sequence simultaneously. In the context of FSLR, this allows the model to capture both spatial relationships (e.g., between hand-shape and facial expression within a frame) and temporal dependencies (e.g., smooth transitions across frames) that are critical for accurately interpreting sign language (Althubiti & Algethami, 2024).

Each attention head will learn its own projection of Queries, Keys, and Values, allowing the model to focus on different sub-features of the input. The outputs of all heads will be aggregated, enabling the model to learn both short- and long-range dependencies across the input sequence (Vaswani et al., 2017; Batnasan, 2024).

After all encoder blocks, the model outputs a sequence of $T \times 768$ frame embeddings. These are aggregated via a learnable CLS-token pooling into a single 1×768 vector. From this pooled vector, two parallel dense layers project $768 \rightarrow 105$ and

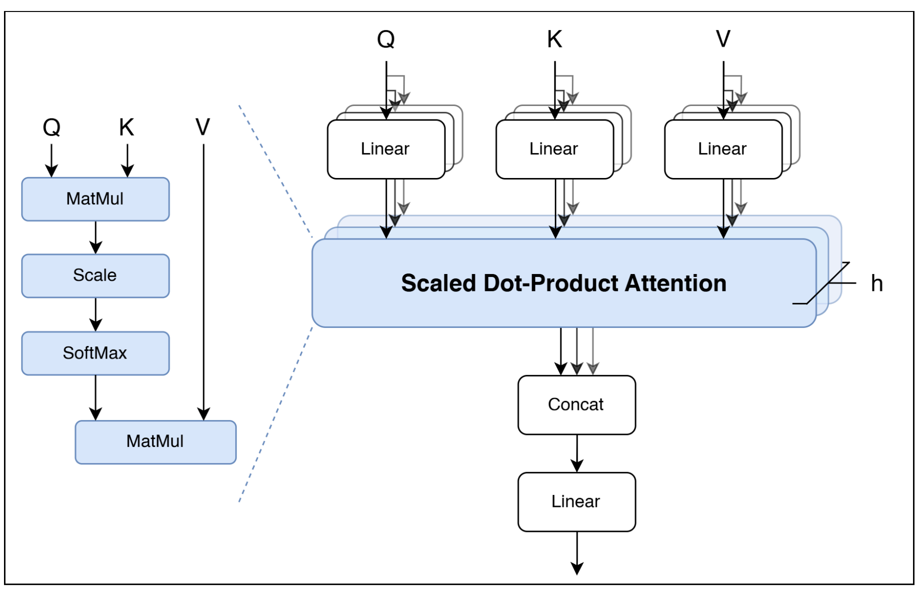
768 → C, each followed by softmax. The first branch produces the gloss prediction (one of 105), and the second produces the category prediction (one of C).

This approach follows the common practice of attaching a feed-forward classifier to Transformer encoders for sequence classification tasks. By using the Transformer's self-attentive encoder to generate a sequence representation and then a dedicated classification head to produce the final prediction, the model will directly output the predicted FSL gloss for each input video.

Overall, this architecture enables modeling of spatio-temporal features across the video input. MHAM dynamically weighs these signals while positional encoding preserves frame order, supporting direct and accurate gloss recognition and semantic category prediction.

Multi-Head Attention Mechanism (MHAM).

Figure 9. Multi-Head Attention Mechanism (MHAM)



The MHAM enables the Transformer to focus on different parts of an input sequence simultaneously (Vaswani et al., 2017). As shown in Figure 9, MHAM extends self-attention by running multiple attention "heads" in parallel, with each head processing the input independently through its own learned projections. By attending to different aspects of the data concurrently (denoted as h parallel Scaled Dot-Product Attentions), the model captures a broader range of relationships within the sequence (Batnasan, 2024).

This structure allows the model to learn information at multiple levels of granularity. Some heads may focus on short-term local patterns, while others attend to long-range dependencies, enabling the representation of both fine-grained and global contextual features (Vaswani et al., 2017; Batnasan, 2024).

In each self-attention unit, input elements (e.g., frame embeddings) are projected into three vectors: Query (Q), Key (K), and Value (V). The Query represents the information an element seeks, the Key describes what is offered by other elements, and the Value contains the information to be aggregated if a match is found. Each Query interacts with all Keys to determine attention weights, with stronger Query-Key matches leading to greater influence during aggregation (Vaswani et al., 2017).

The core operation inside each head is Scaled Dot-Product Attention, expanded on the left side of Figure 9, and proceeds through four steps:

Matrix Multiplication: Queries are multiplied by the transposed Keys to produce raw attention scores.

Scaling: Scores are divided by the square root of the Key dimension to stabilize gradients (Vaswani et al., 2017).

Softmax: The scaled scores are normalized into a probability distribution.

Weighted Sum: The resulting weights are used to compute a weighted sum of the Values, producing the attention output.

Through this mechanism, each input element dynamically aggregates context from the sequence based on learned relevance.

In a basic self-attention mechanism, a single attention head computes all interactions using one set of learned projections for queries, keys, and values. This single-head approach condenses the contextual information into one representation, which can limit the diversity of features captured.

In contrast, MHAM addresses this limitation by employing multiple attention heads (h) in parallel, each with its own learned projections for queries, keys, and values. By attending to different representation subspaces simultaneously, these heads capture a richer and more diverse set of patterns within the input (Vaswani et al., 2017). Each head can specialize in distinct aspects—such as temporal dynamics, spatial structures, or feature subsets (Batnasan, 2024)—and their outputs are concatenated to form a combined representation.

To return to the expected dimensionality and integrate the specialized information, the concatenated outputs are passed through a final Linear projection. This design, along with the use of independent Linear layers before attention, allows each head to focus uniquely while enabling the model to synthesize the heads' outputs into a coherent feature space (Vaswani et al., 2017).

Following the multi-head attention operation, the enriched sequence representations proceed to the next stage in the Transformer Architecture, specifically into the encoder's position-wise feed-forward network for further transformation.

Development Methodology.

The study will follow an Iterative Development Methodology, enabling repeated cycles of model training, evaluation, and refinement throughout the experimentation process. Each iteration will incorporate insights from performance metrics and error analysis, allowing the system to evolve based on empirical results rather than fixed development stages.

To implement this system, the development will be carried out primarily using Python, due to its extensive ecosystem and community support in machine learning and computer vision. The PyTorch framework will serve as the backbone for building and training the Transformer model, offering dynamic computation graphs and fine-grained control over model parameters.

For video and image data processing, OpenCV and NumPy will be utilized to handle frame extraction, format conversions, and numerical operations. To extract pose and keypoint data from each frame, the system will integrate MediaPipe, which provides state-of-the-art landmark detection crucial for recognizing both manual and non-manual sign language cues.

Evaluation of model performance will be supported by Scikit-learn, particularly for computing classification metrics such as F1-score and accuracy. Visualizations will be handled by Matplotlib and Seaborn.

Development and experimentation will take place both in a Vast AI environment, for scalable cloud-based execution, and locally via Jupyter Notebooks, especially during preprocessing and initial testing stages. For writing and organizing code efficiently, Visual Studio Code will be used as the primary IDE.

Lastly, Weights & Biases (WandB) will be integrated for experiment tracking, helping monitor training logs, hyperparameter configurations, and performance metrics across multiple runs—critical for ensuring transparency and reproducibility in model development.

Data Generation Procedure

This section describes the experimental workflow used to compare the MHAM Transformer and IV3-GRU models under varying occlusion conditions. It provides a detailed overview of the preprocessing, training, and evaluation procedures, which are organized into three distinct phases: Pre-Experimentation, Experimentation, and Post-Experimentation.

Pre-Experimentation Phase.

The Pre-Experimentation Phase involves the preparation of data and the training of models, ensuring consistent preprocessing and controlled training conditions.

Keypoint Extraction & Normalization.

The data preparation process begins with extracting keypoints from video frames using the MediaPipe framework. This system captures essential features for Filipino Sign Language (FSL) recognition, including both manual and non-manual components like hand gestures and facial expressions. These keypoints will be normalized to ensure consistency across frames and videos. The data will then be organized into time-ordered sequences, which are crucial for capturing the dynamic nature of sign language. This structured data will form the foundation for model training and evaluation.

Pipeline Compatibility Check.

After preprocessing, the next step is verifying that the data is compatible with the model pipeline. Key factors such as input shapes, padding behavior, and batch processing mechanisms will be checked to ensure they align with the model's requirements. Any inconsistencies or errors in data formatting will be addressed to ensure the data flows smoothly through the pipeline.

Model Training.

Model training will begin with the selection of the MHAM Transformer and IV3-GRU baseline models. Both models will be trained from scratch using the preprocessed data within the PyTorch framework. Identical hyperparameters will be initialized for both models to ensure a fair comparison. Training will occur in two phases: first, the Transformer model will be trained on a small subset of 10 signs from the "Greetings" category as an initial evaluation. Following this, the model's weights will be reinitialized, and the model will be trained on the full 105-sign dataset to assess generalization across a broader set of signs. Adam optimizer will be used to update the model's weights, with batch size and epoch numbers optimized during preliminary tuning.

Define and Apply Stopping Criteria.

To prevent overfitting or unnecessary training, stopping criteria will be established. A predetermined number of epochs (e.g., 50) will be set, with training halting after this point unless earlier termination is triggered by a lack of improvement in the validation loss over consecutive epochs, a condition known as "patience." The delta parameter will be adjusted to balance performance with training efficiency.

Experimentation Phase.

The Experimentation Phase involves running inference on the held-out test (20%) sets and logging model outputs and performance metrics.

Determine If Sign is With or Without Occlusion.

Each test clip is automatically flagged as "with occlusion" if any of the core facial landmarks (eyes, nose, mouth) tracked by MediaPipe disappear from the landmark set U(f) for a temporally contiguous span, indicating a hand–head overlap. To suppress brief tracking glitches, both ground truth and detections are passed through a 5-frame majority filter: an occlusion event begins when a landmark falls out of U(f) and only ends when it remains present again for three out of five consecutive frames (Viitaniemi et al., 2013). Clips not meeting these criteria are labeled "without occlusion."

Inference and Log Predictions.

Once categorized into clean or occluded conditions, both the MHAM Transformer and IV3-GRU models will be evaluated on the respective test sets. The models will generate predictions for both recognition (identifying the correct gloss labels) and classification (categorizing glosses into semantic categories). The predictions, along with their confidence scores, will be recorded and compared to the ground truth labels to measure performance under both clean and occluded conditions.

Aggregate Per-Sample Logs (TP/FP/TN/FN).

For each test sample, the predictions will be logged in structured logs, capturing key metrics such as True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). These metrics will be used to compute macro-averaged precision, recall, and F1-score for both recognition and classification tasks. The logs will ensure that the performance of each model can be thoroughly evaluated at the individual sample level, providing insights into the strengths and weaknesses of each model.

Record Training Metadata.

In addition to logging predictions and performance metrics, metadata related to the training process will be recorded. This includes tracking training and validation loss curves, the convergence epoch (when training stabilizes), final accuracy values, and hyperparameter configurations. These records will ensure transparency and reproducibility of the experiment.

Post-Experimentation Phase.

The Post-Experimentation Phase focuses on analyzing the results obtained from the inference and logging processes. This phase includes computing performance metrics, conducting statistical testing, and evaluating model robustness to occlusion.

Compute descriptive metrics (precision, recall, F_1).

After obtaining the per-sample precision, recall, and F_1 scores for both recognition and classification tasks, the macro-averaged precision, recall, and F_1 scores will be computed for each model under clean and occluded conditions. These macro-averaged metrics will provide a balanced comparison of model performance, ensuring equal weighting for each gloss category regardless of frequency.

Perform paired statistical tests with assumption checks.

To evaluate whether the differences in performance are statistically significant, paired statistical tests will be conducted. The normality of paired differences will first be assessed using the Shapiro-Wilk test. If the differences are normally distributed, a paired t-test will be used; otherwise, the Wilcoxon signed-rank test will be applied. These tests will assess whether the observed performance differences between the two models are statistically significant.

Apply multiple-comparisons correction.

Since multiple statistical tests will be performed across different occlusion levels and performance metrics, the Holm-Bonferroni correction will be applied to adjust for multiple comparisons. This correction ensures that the family-wise error rate is controlled, reducing the likelihood of false positives in statistical testing.

Conduct error analysis on per-sample failures.

Error analysis will be conducted to investigate the misclassifications made by each model. This analysis will focus on the false positives (FP) and false negatives (FN) from the per-sample logs, which were recorded in the Experiment Report Template. By identifying specific glosses or categories that were more likely to be misclassified under occluded versus clean conditions, the error analysis will help pinpoint areas where the models struggle. The process will also compare the performance of the MHAM Transformer and IV3-GRU model, particularly under occlusion, and reveal patterns of misclassification that can be addressed in future model iterations.

Experiment Report Template.

The Experiment Report Template organizes each run into two main sections. The first section comprises four summary tables (Tables 3–6) that present recognition and classification metrics—precision, recall, and F_1—for both the MHAM Transformer and the IV3-GRU with and without occlusion. Each summary table reports both models' scores, the difference Δ and the corresponding H_0 decision.

The second section contains four per-sample log tables (Tables 7–10) that capture individual clip outcomes for each model and task. Each log entry records the video filename, true versus predicted gloss or category, and counts of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). This two-part structure ensures a consistent presentation of raw scores, practical significance, and statistical testing across all experimental conditions.

Table 3

RQ1 – Recognition (Without Occlusion)

Metric	**Transformer Score**	**IV3-GRU Score**	**Δ (T – I)**	**p-Value**	**H_{01} Decision**
Precision	—	—	—	—	Reject / Fail to Reject
Recall	—	—	—	—	Reject / Fail to Reject
F_1-Score	—	—	—	—	Reject / Fail to Reject

Table 4

RQ2 – Recognition (With Occlusion)

Metric	**Transformer Score**	**IV3-GRU Score**	**Δ (T – I)**	**p-Value**	**H_{01} Decision**
Precision	—	—	—	—	Reject / Fail to Reject
Recall	—	—	—	—	Reject / Fail to Reject
F_1-Score	—	—	—	—	Reject / Fail to Reject

Table 5

RQ3 – Classification (Without Occlusion)

Metric	**Transformer Score**	**IV3-GRU Score**	**Δ (T – I)**	**p-Value**	**H_{01} Decision**
Precision	—	—	—	—	Reject / Fail to Reject
Recall	—	—	—	—	Reject / Fail to Reject
F_1-Score	—	—	—	—	Reject / Fail to Reject

Table 6

RQ4 – Classification (With Occlusion)

Metric	**Transformer Score**	**IV3-GRU Score**	**Δ (T – I)**	**p-Value**	**H_{01} Decision**
Precision	—	—	—	—	Reject / Fail to Reject
Recall	—	—	—	—	Reject / Fail to Reject

Continuation of Table 6

Metric	Transformer Score	IV3-GRU Score	Δ (T – I)	p-Value	H_{01} Decision
F_1-Score	—	—	—	—	Reject / Fail to Reject

Table 7

Per-Sample Recognition Log – Transformer

Video File	True Gloss	Predicted Gloss	Correct?	TP	FP	TN	FN
filename_0001.mp4	hello	hello	Y	1	0	104	0
…	…	…	…	…	…	…	…

Table 8

Per-Sample Recognition Log – IV3-GRU

Video File	True Gloss	Predicted Gloss	Correct?	TP	FP	TN	FN
filename_0001.mp4	hello	hello	Y	1	0	104	0
…	…	…	…	…	…	…	…

Table 9

Per-Sample Classification Log – Transformer

Video File	True Category	Predicted Category	Correct?	TP	FP	TN	FN
filename_0001.mp4	greeting	greeting	Y	1	0	9	0
…	…	…	…	…	…	…	…

Table 10

Per-Sample Classification Log – IV3-GRU

Video File	True Category	Predicted Category	Correct?	TP	FP	TN	FN
filename_ 0001.mp4	greeting	greeting	Y	1	0	9	0
…	…	…	…	…	…	…	…

Ethical Considerations

The FSL-105 dataset is publicly available and includes video recordings of FSL signers, all credited for their contributions. It was created with informed consent, ensuring participants understood how their recordings would be used. This study follows the dataset's original guidelines and ensures no additional personally identifiable information (PII) is introduced during processing.

Moreover, FSL-105 is also openly accessible for research, education, and development. Proper attribution will be given to Tupal and his collaborators in all outputs derived from the dataset. To supplement the limited public documentation, we obtained permission from Tupal for full access to his research. His work serves as a primary reference, and we fully adhere to the ethical and usage guidelines he outlined. The dataset carries no restrictive licensing, and its reuse is permitted with appropriate credit.

The dataset may not fully represent the diversity of the Filipino Deaf community due to:

1. Demographic constraints (e.g., age, gender, regional variation)
2. Linguistic gaps (e.g., underrepresented signs)
3. Contextual limitations (e.g., absence of sentence-level signing)

These limitations, acknowledged by the dataset creator, are recognized and considered in this study's application and interpretation of results.

Data Analysis

Following the data generation procedure, this section involves evaluating the models' performance based on the results obtained during the Experimentation Phase. The analysis includes computing descriptive metrics, performing statistical testing, conducting error analysis, and visualizing the results to compare the MHAM Transformer and IV3-GRU models under clean and occluded conditions.

Descriptive Analysis.

After obtaining the per-sample precision, recall, and F_1 scores for both recognition and classification tasks, the macro-averaged precision, recall, and F_1 scores will be computed for each model under both clean and occluded conditions. These computed metrics will enable an overall evaluation of how well each model performs in both recognizing individual glosses and classifying them into semantic categories.

Summarize precision, recall, and F_1 by model and occlusion conditions.

Macro-averaged metrics will be computed per model and condition. These metrics will be calculated as follows:

Table 11

Metrics Formula

Metric	Formula
Precision measures the proportion of correct predictions out of all predicted glosses:	$= \frac{1}{N}\sum_{i=1}^{N}\left(\frac{True\ Positives\ (TP)_i}{True\ Positives\ (TP)_i + False\ Positives\ (FP)_i}\right)$
Recall measures the proportion of correct predictions out of all actual glosses:	$= \frac{1}{N}\sum_{i=1}^{N}\left(\frac{True\ Positives\ (TP)_i}{True\ Positives\ (TP)_i + False\ Negatives\ (FN)_i}\right)$
F1-score is the harmonic mean of precision and recall, providing a balanced measure of model performance:	$= \frac{1}{N}\sum_{i=1}^{N}\left(2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i}\right)$

These metrics will be macro-averaged across all classes to ensure that each gloss category is given equal weight, preventing dominant categories from skewing the results.

Tabulate raw metric values and differences.

For each model and condition, raw values for precision, recall, and F_1-score will be recorded. The differences in performance between the MHAM Transformer and IV3-GRU models will be computed, highlighting the impact of occlusion on each model’s ability to recognize and classify signs. The performance difference for each sample is computed as:

$$D_i = M_{Transformer,\, i} \; - \; M_{IV3-GRU,}\, i$$

Where D_i is the difference for sample i, and M denotes the metric (e.g., F1-score).

Statistical Testing.

To assess whether the differences in performance are statistically significant, paired statistical tests will be conducted. The approach for testing normality and selecting the appropriate statistical test is outlined below:

Check normality (Shapiro–Wilk).

The Shapiro-Wilk test will be used to assess the normality of the paired differences between the models' performance metrics. This test will determine whether the differences follow a normal distribution. The formula for the Shapiro-Wilk test statistic is as follows:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2}$$

Where $x_{(i)}$ is the ordered sample value, $\bar{x}$ is the sample mean, and a_i are constants derived from the covariance matrix of the order statistics.

Select and run paired t-test or Wilcoxon.

The choice of statistical test depends on the distribution of the differences. If the differences are normally distributed, a paired t-test will be applied to compare the means of the paired differences. Otherwise, the Wilcoxon signed-rank test will be used instead, which is a non-parametric alternative.

Paired t-Test for Normal Data.

The paired t-test compares the means of the paired differences between the two models across each performance metric to determine if the means are significantly different from zero.

1. Calculate the paired differences between model metrics:

$$d_i = X_{Transformer,\, i} - X_{IV3-GRU,\, i}$$

   Where:

   a. $X_{Transformer,\, i}$ and $X_{IV3-GRU,\, i}$ are the performance metrics of the Transformer model and IV3-GRU model

2. Compute the mean and standard deviation of the differences using the formula:

   a. Mean: $\overline{d} = \frac{\sum_{i=1}^{n} d_i}{n}$

      Where:

      a. d_i represents each individual difference between the paired observations

      b. n is the number of pairs of observations

      c. $\sum_{i=1}^{n} x_i$ is the sum of all the differences

   b. Standard Deviation: $s_d = \sqrt{\frac{\sum_{i=1}^{n} \left(d_i - \overline{d}\right)^2}{n-1}}$

Where:

a. d_i is each individual difference between the paired observations

b. $\overline{d}$ is the mean differences

c. n is the number of values in the dataset

d. $\sum_{i=1}^{n}\left(d_i - \overline{d}\right)^2$ is the sum of the squared deviations of each difference from the mean of the differences

3. Apply the t-test formula:

$$t = \frac{\overline{d}}{\frac{s_d}{\sqrt{n}}}$$

Where:

a. $\overline{d}$ is the mean of paired differences

b. s_d is the standard deviation of the differences

c. n is the number of samples

4. Compare the t-value with the critical value from the distribution table.

Wilcoxon Signed-Rank Test for Non-normal Data.

For non-normal differences, the Wilcoxon signed-rank test will be applied. This ranks the absolute differences and assesses whether the median difference is significantly different from zero.

1. Calculate absolute differences between paired observations (i.e., subtract the second value from the first for each pair).

$$|d_i| = |x_i - y_i|$$

Where:

a. x_i and y_i are the paired values

2. Rank the absolute differences in ascending order. If there are tied values, assign the average rank to each tied value.
3. Assign the signs of the ranks to the differences. If the difference of d_i is positive, the rank will be positive. If the difference d_i is negative, the rank will be negative.
4. Sum the ranks of the positive differences (W^+) and the ranks of the negative differences (W^-).
5. Calculate test statistic W, which is smaller of the two sums: W^+ and W^-.

$$W = min\,(W^+,\ W^-)$$

6. Determine the critical value from the Wilcoxon Signed-Rank distribution table based on the sample size n (number of pairs), and the significance level $\alpha = 0.05$.
7. Compare the test statistic to the critical value from the Wilcoxon table.
   a. If W is less than or equal to the critical value, reject the null hypothesis (there is a significant difference between the pairs).
   b. If W is greater than the critical value, then fail to reject the null hypothesis (there is no significant difference between the pairs).

Correct for Multiple Comparisons.

Since multiple statistical tests will be performed (across different metrics and occlusion levels), the Holm-Bonferroni correction will be applied. This method adjusts the significance threshold for each test to control the family-wise error rate, ensuring that the likelihood of committing a Type I error is minimized. The following steps will be followed:

1. Arrange the p-values in ascending order:

   The p-values from all the hypothesis tests will be sorted in ascending order, so that:

   $$p_1 \leq p_2 \leq \ . \ . \ . \ \leq \ p_m$$

   Where p_1 is the smallest p-value, p_2 is the second smallest, and so on.

2. Calculate adjusted p-values:

   For each p-value p_i (where $i = 1, 2 ..., m$), calculate the adjusted p-value p_i^{adj} using the following formula:

   $$p_i^{adj} = p_i \times (m - i + 1)$$

   This formula adjusts the p-value by multiplying it by the number of tests remaining, accounting for the increasing stringency as more tests are performed.

3. Compare adjusted p-values with significance level α:

   The null hypothesis for each test is rejected if the adjusted p-value p_i^{adj} is less than the significance level $\alpha = 0.05$. Formally, reject H_0 for test i if:

   $$p_i^{adj} \leq \alpha\,(0.05)$$

   If the adjusted p-value p_i^{adj} exceeds α, do not reject the null hypothesis for that test and all subsequent tests.

Report exact p-values and H_0 decision.

The final step in the statistical testing process involves reporting the exact p-values for each hypothesis test, along with the decision regarding the null hypothesis (H_0). If the adjusted p-value is less than 0.05, the null hypothesis will be rejected, indicating that there is a statistically significant difference between the models' performances. If the adjusted p-value exceeds 0.05, the null hypothesis will fail to be rejected, suggesting no significant difference between the two models.

All results, including the p-values and decisions, will be clearly documented and presented to ensure transparency and reproducibility in the analysis.

Error Analysis.

Error analysis is critical for identifying common failure patterns across the MHAM Transformer and IV3-GRU models, particularly in relation to misclassified samples. This analysis will focus on understanding the influence of occlusion on the models' performance, with a particular emphasis on False Positives (FP) and False Negatives (FN).

Impact of Occlusion

The first step in error analysis will be to identify and categorize samples into False Positives and False Negatives. This categorization will facilitate an exploration of which glosses or categories are most susceptible to misclassification under condition with and without occlusion. The analysis will assess how occlusion affects patterns and whether occlusion results in increased

FP or FN. It will also explore which signs are more prone to performance degradation when certain keypoints (such as facial expressions or hand positions) are occluded.

False Positives and False Negatives Analysis

Following the initial categorization of samples, a more detailed analysis will be conducted on False Positives and False Negatives. For False Positives, the analysis will identify whether specific signs are misclassified more frequently as one class when they do not belong to it, particularly under occluded conditions. Signs with visually similar features or gestures may be prone to misclassification due to occluded keypoints, leading to confusion in the classification process.

For False Negatives, the focus will shift to determining whether occluded conditions increase the number of missed classifications, where the model fails to recognize signs that should have been correctly identified. This is especially relevant for signs that rely on subtle facial expressions or intricate hand movements, which may be more challenging to detect when portions of the keypoints are occluded. The analysis will uncover which signs are more likely to be overlooked by both models, particularly under occluded conditions, providing insights into the specific weaknesses of each model in handling occlusion-induced challenges.

Visualization.

Visualization will play a critical role in presenting and interpreting the results of the analysis. Plots and charts may be created to provide a clear understanding of how key performance metrics behave across clean and occluded conditions.

Plot metrics across occlusion levels.

Line plots may be generated to visually demonstrate how key performance metrics—precision, recall, and F_1 score—change across clean and occluded conditions. These metrics will be plotted for both the MHAM Transformer and IV3-GRU models. Separate plots may also be generated for the recognition and classification tasks, as these metrics may behave differently depending on the nature of the task. By examining these line plots, trends in model performance across the different conditions will be revealed. This will highlight which model is more resilient to occlusion and where performance degradation becomes more pronounced, allowing for the identification of threshold points at which occlusion begins to significantly affect model accuracy.

Chart confidence indicators.

Bar charts may be created to visualize the performance differences between the MHAM Transformer and IV3-GRU models for each metric (precision, recall, and F_1 score) under clean and occluded conditions. Each bar will represent the difference in performance between the two models for clean versus occluded conditions.

To provide additional clarity, confidence intervals may also be included in the charts for each performance metric. These intervals will give a range within which the true value of the metric is likely to fall, offering a sense of the reliability and stability of the observed differences between the models. The inclusion of confidence intervals will help assess whether the performance differences are statistically significant and whether these differences remain consistent across the tested conditions.

Table 12

Mapping of Research Questions to Data and Methods

Research Question	Data Sources	Data Analysis
Central Question		
How does a Multi-Head Attention Mechanism Transformer improve recognition and classification of isolated FSL glosses compared to the IV3-GRU baseline, both with and without occlusion?	Derived from RQ1, RQ2, RQ3, and RQ4	Summary of results from RQ1–4: Comparative analysis of MHAM Transformer and IV3-GRU performance based on recognition and classification metrics, including statistical significance testing.
Sub-Question 1		
1. RQ1: What is the performance of IV3-GRU and MHAM Transformer in recognizing sign glosses without occlusion in terms of Precision, Recall, and F1-Score?	Test dataset of isolated FSL glosses with ground truth labels, Gloss-level prediction outputs from MHAM Transformer and IV3-GRU	Compute per-gloss: Precision, Recall, F1-score. Aggregate macro-averaged scores per model. Analyze confusion matrix (TP, FP, FN). Log gloss-level results.
Sub-Question 2		
2. RQ2: What is the performance of IV3-GRU and MHAM Transformer in recognizing sign glosses with occlusion in terms of Precision, Recall, and F1-Score?	Occluded test set of isolated FSL glosses with ground truth labels, Gloss-level prediction outputs from MHAM Transformer and IV3-GRU	Compute per-gloss: Precision, Recall, F1-score. Aggregate macro-averaged scores per model. Analyze confusion matrix (TP, FP, FN). Log gloss-level results.
Sub-Question 3		
3. RQ3: What is the performance of IV3-GRU and MHAM Transformer in classifying sign glosses without occlusion in terms of Precision, Recall, and F1-Score?	Test dataset of isolated FSL glosses with semantic category labels (e.g., greetings, survival, numbers), Category-level prediction outputs from MHAM Transformer and IV3-GRU	Compute per-gloss: Precision, Recall, F1-score. Aggregate macro-averaged scores per model. Analyze confusion matrix (TP, FP, FN). Log gloss-level results.

Continuation of Table 14

Research Question	Data Sources	Data Analysis
Sub-Question 4		
4. What is the performance of IV3-GRU and MHAM Transformer in classifying sign glosses with occlusion in terms of Precision, Recall, and F1-Score?	Occluded test set of isolated FSL glosses with semantic category labels (e.g., greetings, survival, numbers), Category-level prediction outputs from MHAM Transformer and IV3-GRU	Compute per-category: Precision, Recall, F1-score. Aggregate macro-averaged scores per model. Analyze confusion matrix (TP, FP, FN). Log category-level results.

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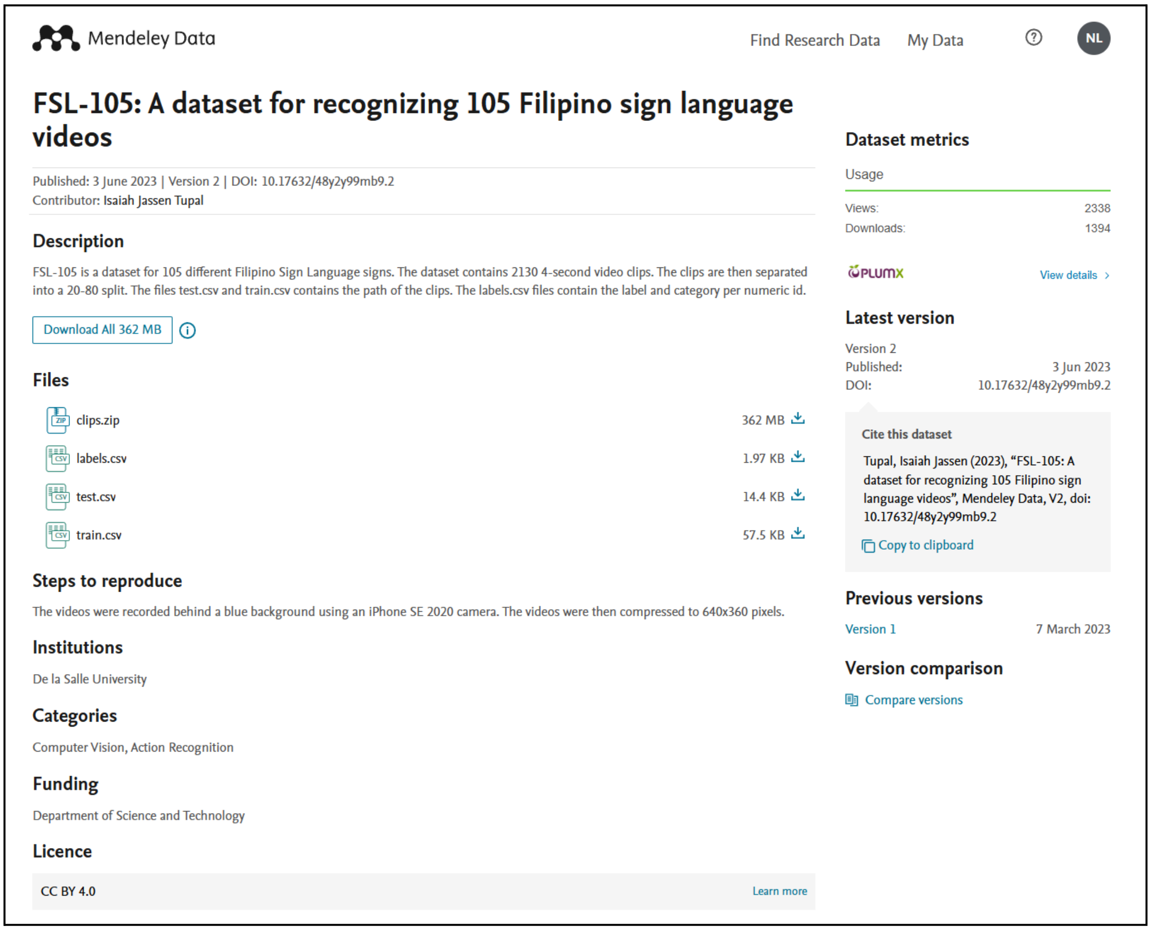
APPENDICES

Appendix 1: FSL-105 Dataset (Raw Video Input)

Figure 10. FSL-105 Dataset Retrieved from Mendeley Data



Mendeley Data   Find Research Data   My Data   NL

FSL-105: A dataset for recognizing 105 Filipino sign language videos

Published: 3 June 2023 | Version 2 | DOI: 10.17632/48y2y99mb9.2
Contributor: Isaiah Jassen Tupal

Description

FSL-105 is a dataset for 105 different Filipino Sign Language signs. The dataset contains 2130 4-second video clips. The clips are then separated into a 20-80 split. The files test.csv and train.csv contains the path of the clips. The labels.csv files contain the label and category per numeric id.

Download All 362 MB

Files

clips.zip	362 MB
labels.csv	1.97 KB
test.csv	14.4 KB
train.csv	57.5 KB

Steps to reproduce

The videos were recorded behind a blue background using an iPhone SE 2020 camera. The videos were then compressed to 640x360 pixels.

Institutions

De la Salle University

Categories

Computer Vision, Action Recognition

Funding

Department of Science and Technology

Licence

CC BY 4.0   Learn more

Dataset metrics

Usage

Views: 2338
Downloads: 1394

PLUMX   View details

Latest version

Version 2
Published: 3 Jun 2023
DOI: 10.17632/48y2y99mb9.2

Cite this dataset

Tupal, Isaiah Jassen (2023), "FSL-105: A dataset for recognizing 105 Filipino sign language videos", Mendeley Data, V2, doi: 10.17632/48y2y99mb9.2

Copy to clipboard

Previous versions

Version 1   7 March 2023

Version comparison

Compare versions

Appendix 2: Research Instrument – Screen Mockup

Figure 11. File Upload Interface - No File Selected

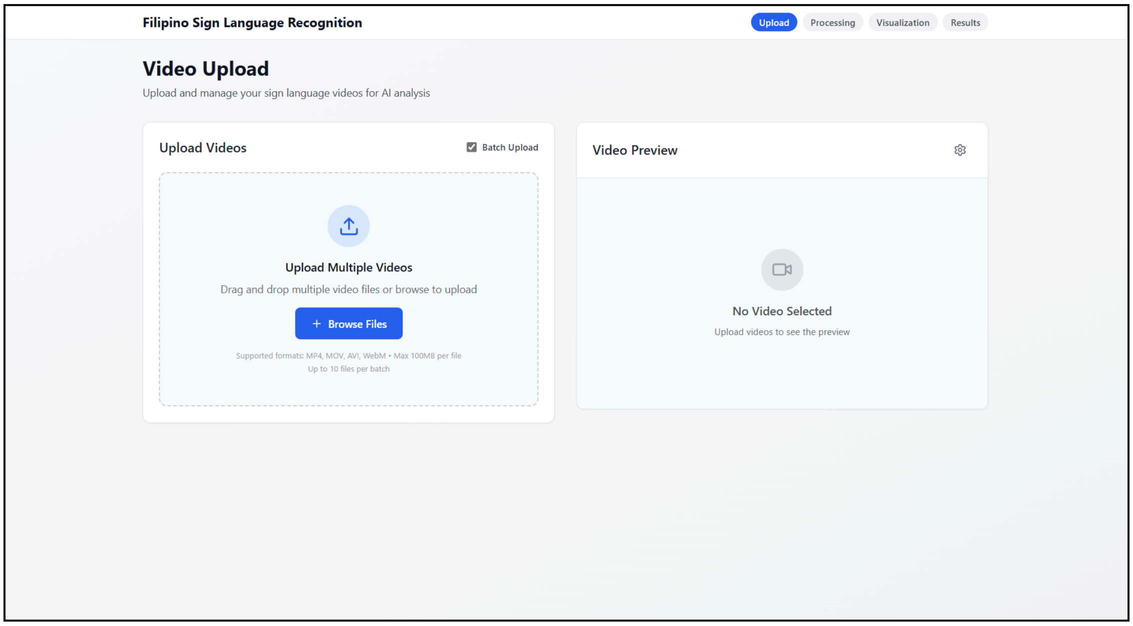


Figure 12. File Upload Interface - Batch Files Preview

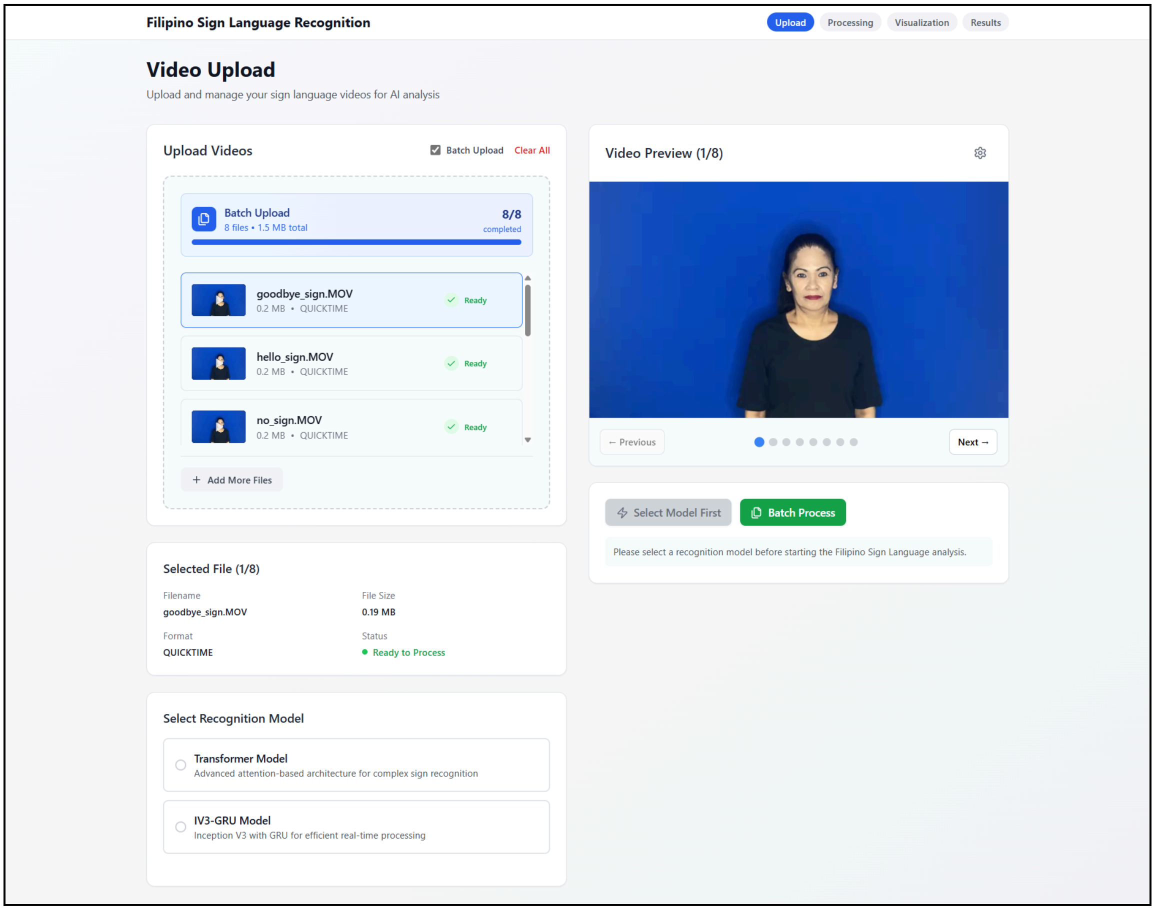


Figure 13. File Upload Interface - Select Transformer Model

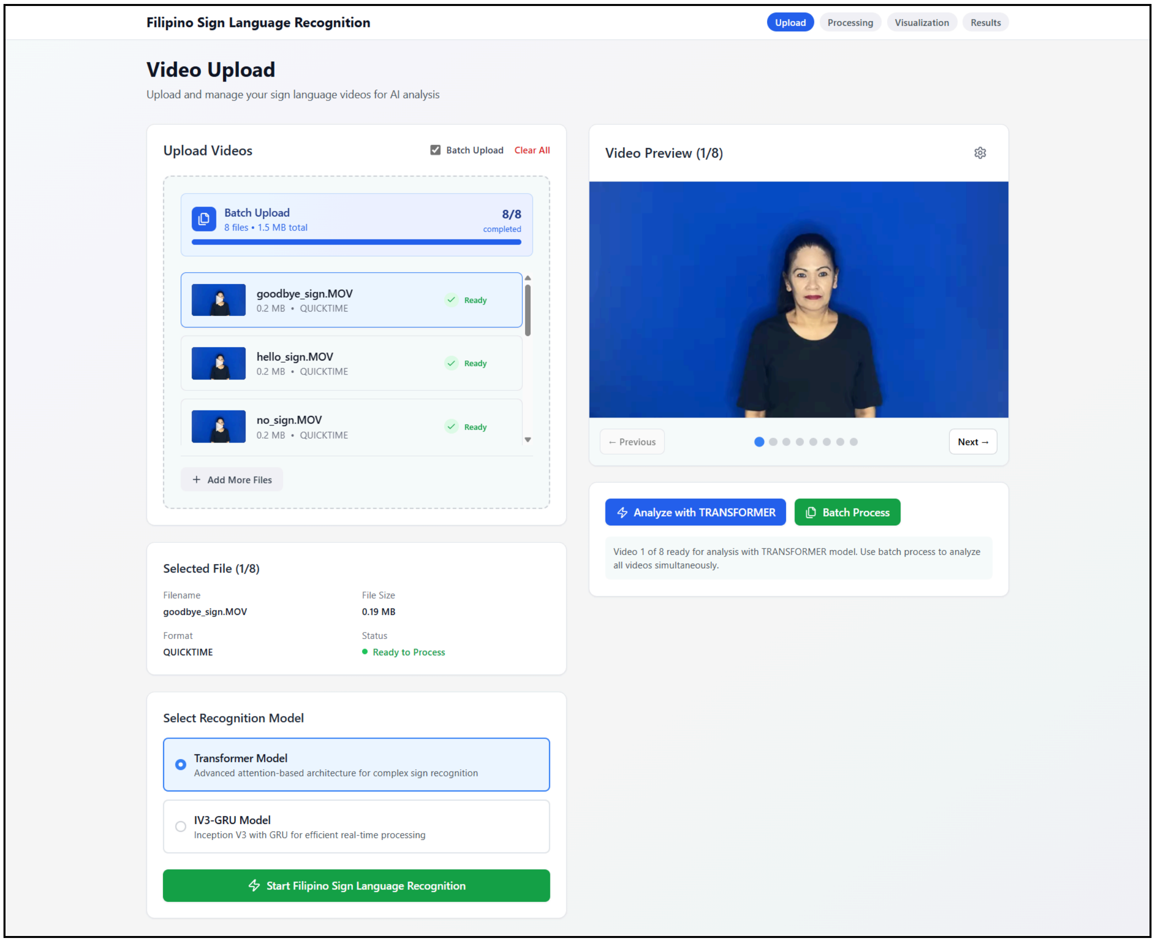


Figure 14. File Upload Interface - Select IV3-GRU Model

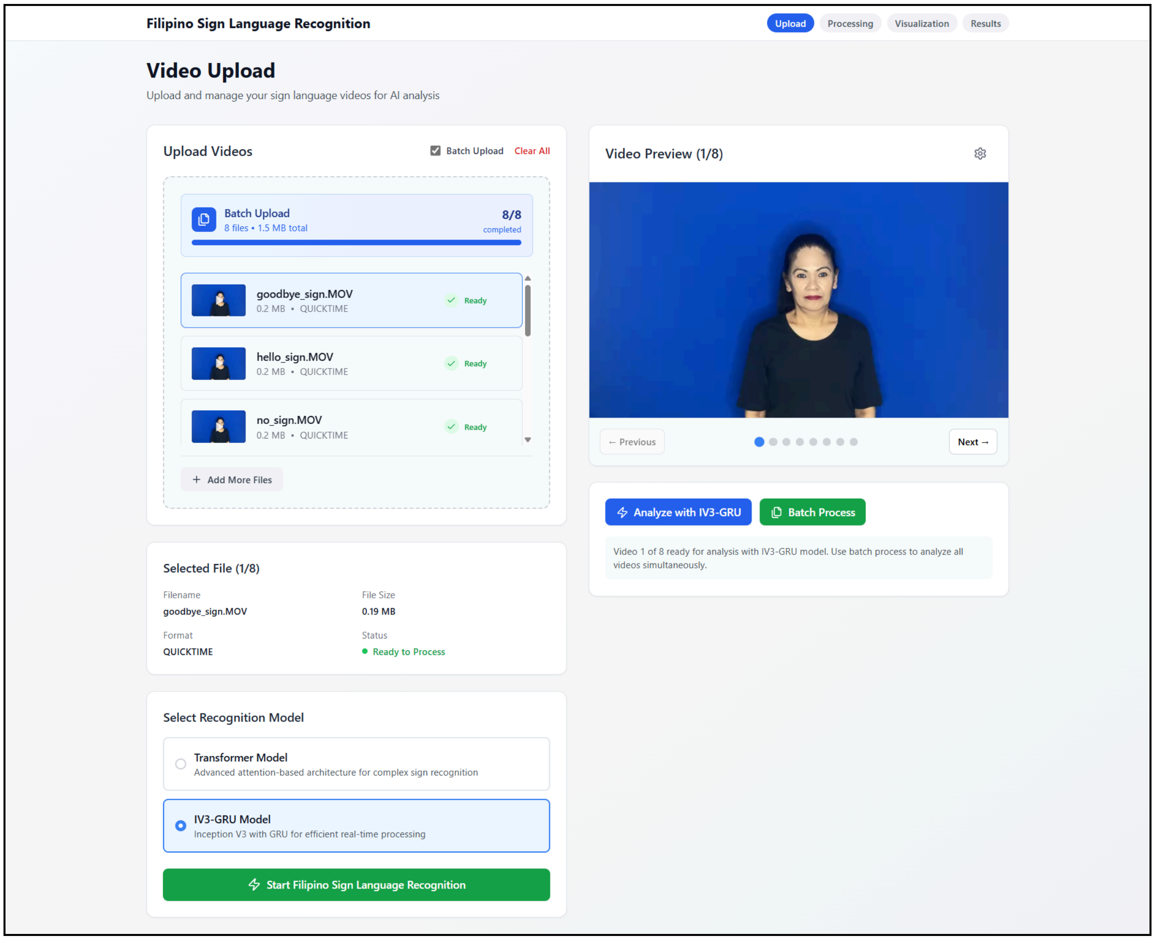


Figure 15. Processing Interface - Transformer Model

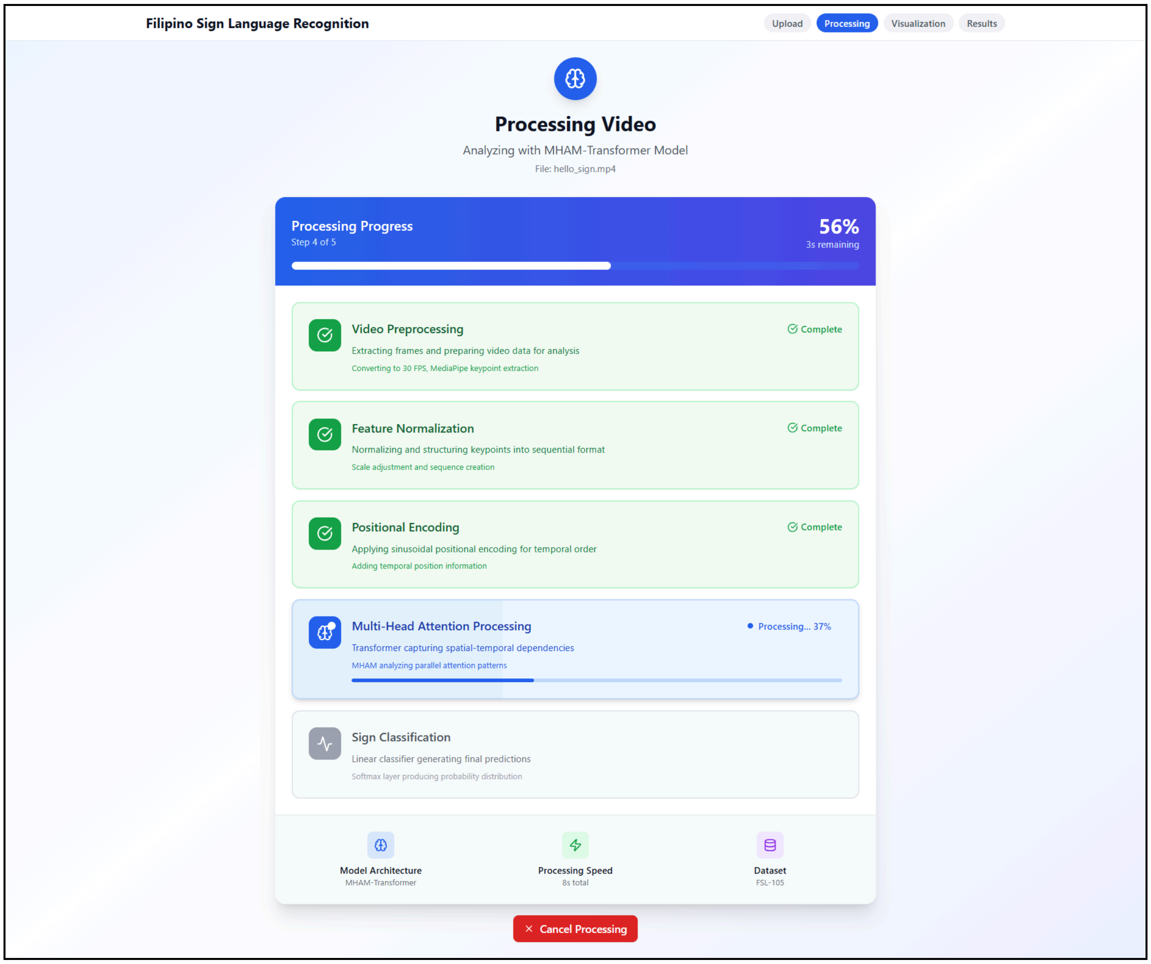


Figure 16. Processing Interface - IV3-GRU Model

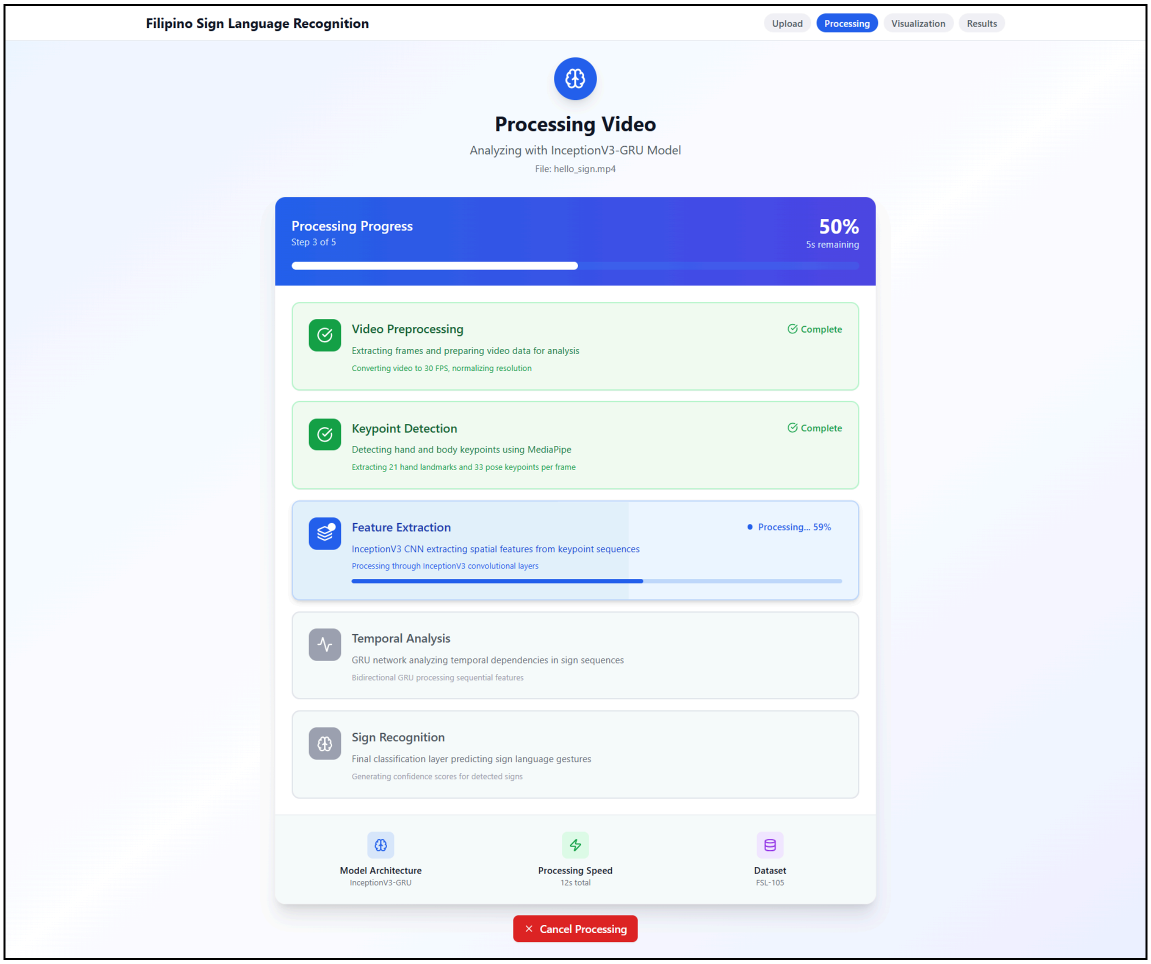


Figure 17. Visualization Interface - Transformer Model

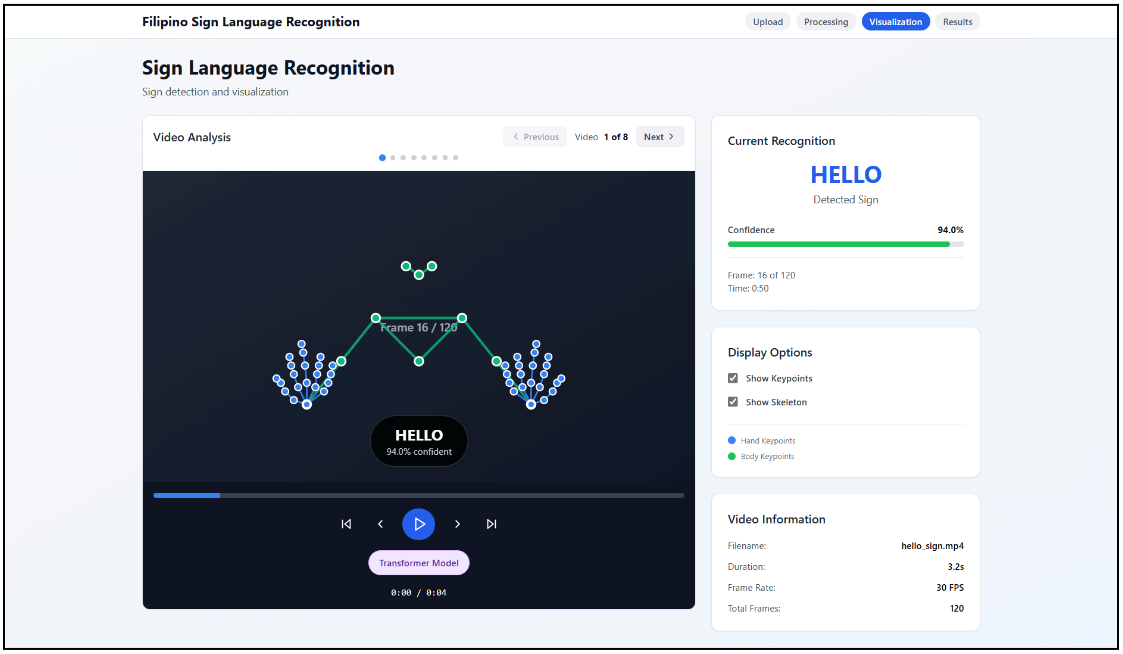


Figure 18. Visualization Interface - IV3-GRU Model

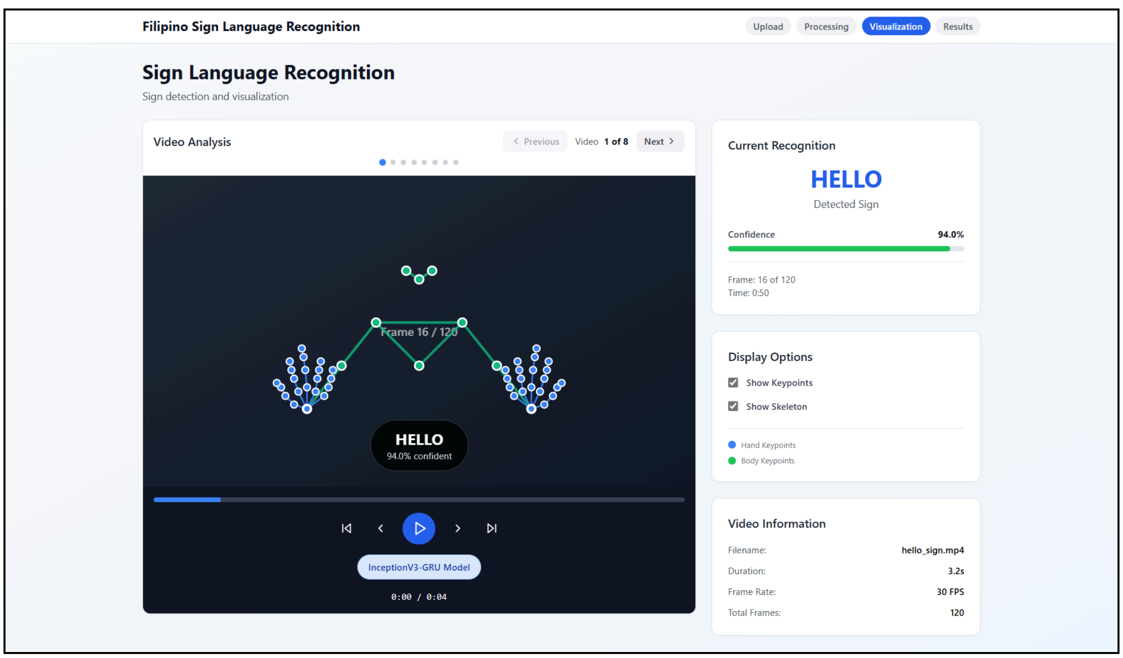


Figure 19. Results Interface - Transformer Model

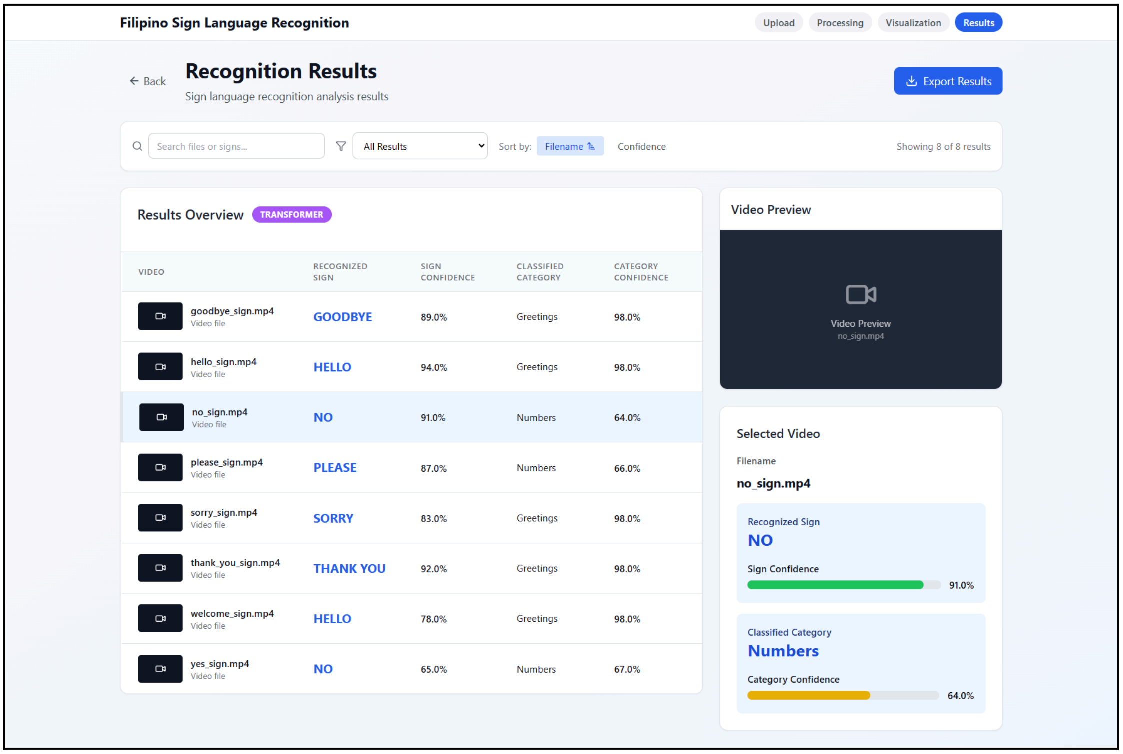


Figure 20. Results Interface - IV3-GRU Model